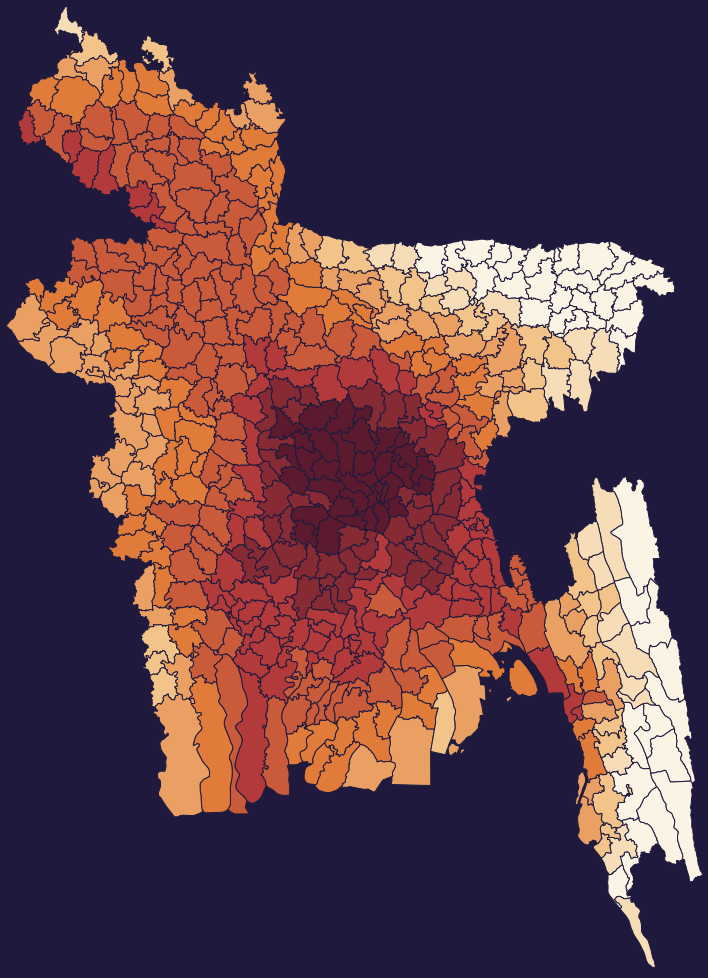
PORTFOLIO PROJECT · GEOSPATIAL ANALYSIS 1998-2024

ANALYSIS REPORT

Bangladesh Air Pollution Exposure Analysis

Population-weighted satellite PM2.5 for every division, district and upazila, with trends, seasonality, spatial clusters and drivers.



Upazila PM2.5, 2020-2024 (µg/m³)

<45 60 >80

67.6 µg/m³
Average person, 2020-2024, 13.5x the WHO guideline

99.4%
of 168 million people above the national standard in 2024

+1.08
µg/m³ a year, Sen's slope 1998-2024

6.1 yrs
life expectancy lost, AQLI estimate

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October 2026
Python · R · DuckDB · Power BI

Contents

Executive summary ii

1. Introduction 1

1.1 Why PM2.5 1
1.2 Questions 1
1.3 Scope 1

2. Data 2

2.1 Sources 2
2.2 Processing 2
2.3 Quality checks 3
2.4 The 2018 spike 3

3. Methods 4

3.1 Trends 4
3.2 Seasonality 4
3.3 Spatial statistics 4
3.4 Spatial regression 5
3.5 Machine learning 5
3.6 Exposure bands and life expectancy 5

4. How much PM2.5 people breathe 6

4.1 Divisions 6
4.2 Districts and upazilas 7
4.3 Where people live matters 9

5. People above the WHO targets 11

5.1 How the distribution moved 11

6. Trends since 1998 13

6.1 A rise and then a plateau 13
6.2 Where it rose fastest 14

7. Seasonality 16

7.1 Which months rose 17
7.2 The seasonal swing is narrowing 17

8. Spatial clustering 20

8.1 How far it reaches 20
8.2 Hot and cold clusters 21

9. Spatial regression 23

9.1 Which model to read 23
9.2 Effects 24

10. Machine learning drivers 26

10.1 Random against spatial cross-validation 26
10.2 What the model leans on 27

11. What cleaner air would be worth 29

12. Power BI dashboard 31

13. Limitations 34

14. Conclusion 35

References 36

Appendix A. National yearly series 37

Appendix B. Files and reproducibility 37

Executive summary

This report measures how much fine particulate matter (PM2.5) people in Bangladesh breathe, where it is worst, how it changed from 1998 to 2024, and what goes with higher levels. It uses the ACAG SatPM2.5 V6.GL.03 satellite-based estimates at about 1 km, weighted by the WorldPop 2022 population, for the whole country, 8 divisions, 64 districts and 507 upazilas. The analysis was done in Python (raster processing, DuckDB SQL, machine learning) and R (trend tests, spatial statistics, spatial regression), and the results are shown in a Power BI dashboard.

67.6	13.5x	99.4%	6.1 years
µg/m³, average person, 2020-2024	the WHO guideline of 5 µg/m³	of people above 35 µg/m³ in 2024	life expectancy lost, AQLI estimate

Main findings

- **Exposure is very high everywhere.** The population-weighted mean rose from 46.1 µg/m³ in 1998-2002 to 67.6 µg/m³ in 2020-2024, an increase of 21.6 µg/m³ (47%). Every person lives above the WHO guideline in every year. In 2024, 167.3 million people (99.4%) lived above 35 µg/m³, the WHO interim target 1 and the Bangladesh national standard. 89% lived above 50 and 25% above 75.
- **The rise stopped around 2010.** The national Sen's slope over 1998-2024 is +1.08 µg/m³ per year (95% CI 0.65 to 1.41, Hamed-Rao corrected Mann-Kendall p 1.7e-05). It was +2.15 per year in 1998-2010 and +0.06 per year (p 0.83) in 2011-2024. 500 of 507 upazilas still show a significant rise over the full period.
- **Dhaka is the centre of the problem.** Dhaka division averaged 79.8 µg/m³ in 2020-2024 and Sylhet 44.8. Dhaka district (83.9) and Savar upazila (85.3) are the worst areas. No division changed its rank between the first and last five years.
- **Winter drives exposure.** January averages 121 µg/m³ and July 26 µg/m³. December rose the most in absolute terms (+2.15 µg/m³ per year) while August rose the most in relative terms (+46% per decade), so the winter to monsoon ratio narrowed from 5.6x to 4.6x.
- **Pollution is strongly clustered.** Moran's I is 0.95 for the 2020-2024 level and 0.91 for the trend. A hot cluster of 117 upazilas around Dhaka holds 53.9 million people. Clustering fades by about 124 km.
- **Distance to Dhaka is the strongest correlate.** In the spatial error model, PM2.5 falls by 7.9 µg/m³ per standard deviation (74 km) of distance from Dhaka. Rainfall and dry-season wind look important in OLS but shrink to -1.1 and +0.4 once spatial dependence is modelled.
- **Random cross-validation overstates ML skill.** XGBoost reached R^2 0.94 with random folds and 0.66 when whole regions were held out. The spatial score is the honest one.
- **The health cost is large.** Using the AQLI method, the average person loses an estimated 6.1 years of life expectancy compared with the WHO guideline (Dhaka division 7.3, Sylhet 3.9). Meeting only the national standard would win back about 3.2 years. Reaching 35 µg/m³ needs an average cut of 46%.

These are modelled estimates of outdoor exposure, not monitor readings, and the regression and SHAP results describe places, not causes. Chapter 13 lists the limitations.

CHAPTER 1

Introduction

1.1 Why PM2.5

PM2.5 is particulate matter smaller than 2.5 micrometres. It goes deep into the lungs and the blood, and long-term exposure is linked to heart disease, stroke, lung cancer and chronic respiratory disease. The WHO lowered its annual guideline to 5 µg/m³ in 2021 and kept four interim targets (35, 25, 15 and 10 µg/m³) as steps towards it. Bangladesh's own annual standard, under the Air Pollution (Control) Rules 2022, is 35 µg/m³, the same as the first interim target.

Dhaka often appears in city rankings of air quality, but those rankings come from a few monitors and say little about the rest of the country. Satellite-based estimates cover every square kilometre, so they can show exposure for all 507 upazilas and for every year since 1998.

1.2 Questions

The project was planned around seven public questions. Each question has one method and one chapter.

Table 1. Questions, methods and chapters.

#	Question	Method	Chapter
Q1	How much PM2.5 does the average person breathe, nationally and in each division, district and upazila?	Population-weighted means of the 0.01° grid	4
Q2	How many people live above the WHO guideline and each interim target?	People by exposure band, percentiles	5
Q3	How has exposure changed since 1998, and where fastest?	Mann-Kendall test, Sen's slope	6
Q4	Which months are worst, and does the seasonal pattern differ by region?	Monthly means, STL decomposition	7
Q5	Is pollution spatially clustered, and where are the hotspots?	Moran's I, LISA, Getis-Ord Gi*	8
Q6	What area characteristics go with higher PM2.5?	OLS and spatial regression	9
Q7	Does machine learning find the same drivers when tested honestly?	Random forest, XGBoost, spatial CV, SHAP	10

1.3 Scope

The unit of analysis is the area (country, division, district, upazila) by year or month, from 1998 to 2024. The project measures exposure and the number of people exposed. It does not estimate deaths or disease from health records. Chapter 11 gives a life expectancy estimate using a published exposure-response relationship (AQLI), and it is labelled as an estimate throughout.

CHAPTER 2

Data

2.1 Sources

Table 2. Data sources.

Data	Source	Detail	Used for
PM2.5, annual	ACAG SatPM2.5 V6.GL.03 (Washington University)	0.01° (about 1 km), 1998-2024, public AWS bucket	All exposure results
PM2.5, monthly	ACAG SatPM2.5 V6.GL.03	0.1° (about 10 km), 1998-2024	Seasonality
Population	WorldPop 2022, constrained	100 m grid, 168.3 million people	Weights for every mean and count
Boundaries	HDX cod-ab-bgd (BBS)	8 divisions, 64 districts, 507 upazilas, pcode keys	Zonal statistics, maps
Elevation	Copernicus GLO-90 DEM	90 m, averaged per upazila	Covariate
Roads	OpenStreetMap (Geofabrik extract)	Motorway, trunk, primary, secondary	Covariate (km per km²)
Wind and rainfall	NASA POWER	About 0.5°, long-term monthly means	Covariates (Dec-Feb wind, annual rain)

One data version (V6.GL.03) is used for all 27 years, because mixing releases creates false jumps in a trend. Every download is logged with its URL, size and date in *data/raw/download_manifest.csv*.

2.2 Processing

Each global PM2.5 file was cropped to Bangladesh straight after download. WorldPop was summed into the 0.01° PM2.5 grid, so every grid cell carries its 2022 population. Each cell was assigned to the upazila its centre falls in, and the 21,135 people (0.013%) whose cells had no upazila centre were moved to the nearest upazila (at most 3.2 km). For each area and year the population-weighted mean is

$$PM2.5_{area} = \Sigma\,(PM2.5_{cell} \times people_{cell}) \,/\, \Sigma\, people_{cell}$$

The area-weighted mean, which ignores where people live, was kept for comparison. Districts and divisions were computed from the cells directly, not by averaging upazila values.

2.3 Quality checks

Table 3. Quality checks from notebook 02.

Check	Result
Cells inside Bangladesh (0.01°)	129,302
Population on the grid	168,343,283 (same in every year)
Missing PM2.5 cells inside the border, any year	0.0%
Upazilas with no 0.01° cell centre	0
People moved to the nearest upazila	21,135 (0.013%), at most 3.2 km
R script 09 vs notebook 02, national yearly mean	identical
R script 09 vs notebook 02, upazila 2024 mean	at most 0.30 µg/m³ apart (cell assignment)

2.4 The 2018 spike

In V6.GL.03 the national 2018 value (83.3 µg/m³) is about 28% above the neighbouring years, in both the annual 0.01° and the monthly 0.1° files. In the previous release (V6.GL.02.04) the same jump was only 3%. The spike is therefore specific to this release and not a real event. It was kept, flagged on every time series figure, and the trends use Sen's slope, which is not pulled by one outlier. Without 2018 the national Sen's slope changes from 1.08 to 1.07 µg/m³ per year.

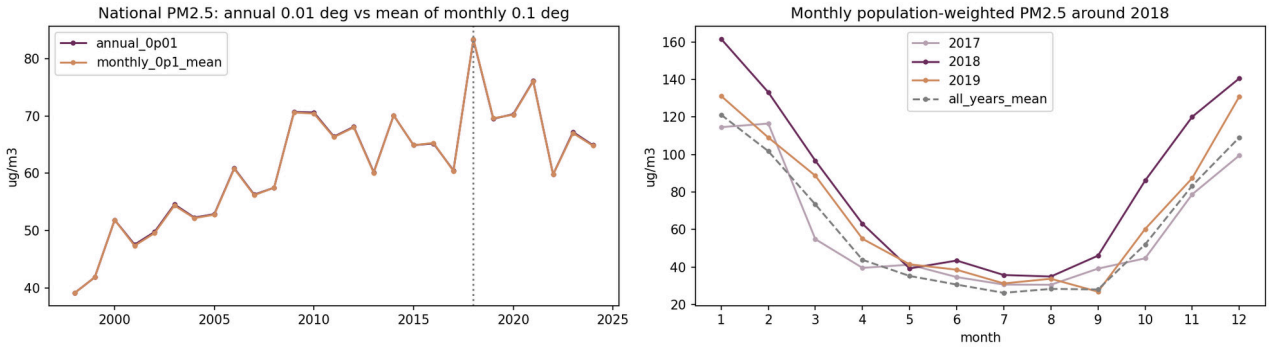


Figure 1. National PM2.5 around 2018. Left: annual 0.01° values against the mean of the monthly 0.1° values. Right: monthly values for 2017, 2018 and 2019 against the all-years mean.

CHAPTER 3

Methods

The work was split by tool, the same way as in my earlier projects: Python notebooks in Google Colab for data, rasters, SQL, exploration and machine learning, then plain R scripts in RStudio for the statistics and the dashboard export.

Table 4. Analysis pipeline.

File	Tool	What it does
01_data_collection.ipynb	Python (requests, xarray)	Download and crop PM2.5, WorldPop and boundaries
02_raster_processing.ipynb	xarray, rioxarray, geopandas	Population weighting, area panels, WHO bands, 2018 check
03_duckdb_sql_analysis.ipynb	DuckDB	12 SQL queries: rankings, change, bands, seasons, weighting gap
04_exploratory_analysis.ipynb	matplotlib	Maps, trend lines, heatmaps, distributions
05_ml_drivers.ipynb	scikit-learn, XGBoost, SHAP	Covariates, random vs spatial CV, SHAP
06_trends_seasonality.R	trend, modifiedmk	Mann-Kendall, Sen's slope, STL, seasonal swing
07_spatial_autocorrelation.R	sf, spdep	Moran's I, correlogram, LISA, Gi*
08_spatial_regression.R	spatialreg, lmtest	OLS, LM tests, SAR, SEM, SLX, SDM, SDEM
09_who_exposure.R	dplyr, sf	Bands, percentiles, cut needed, AQLI years
10_powerbi_export.R	readr	27 tables for the dashboard

3.1 Trends

For every area the yearly series (n = 27) was tested with the Mann-Kendall test using the Hamed-Rao correction for autocorrelation, and the size of the trend is Sen's slope, the median of all pairwise slopes, with a 95% confidence interval. P-values across the 507 upazilas were adjusted with the Benjamini-Hochberg method. To test whether the rise slowed down, the slope was estimated separately for 1998-2010 and 2011-2024. The relative rise is ten times Sen's slope divided by the 1998-2002 mean (% per decade).

3.2 Seasonality

Monthly series were decomposed with STL on the log scale (seasonal window 13), so the seasonal component is a ratio. The seasonal swing is the peak month divided by the trough month of that component, and its yearly values were tested for a trend. Sen's slope was also estimated for each calendar month.

3.3 Spatial statistics

Spatial weights use queen contiguity (upazilas sharing a border), row-standardised, with 6 nearest neighbours as a check. Global Moran's I was tested with 999 permutations. A correlogram over neighbour orders 1 to 10 shows how far clustering reaches. Local clusters use LISA (local Moran's I, p

< 0.05) and Getis-Ord Gi* (90, 95 and 99% classes). Both were run on the 2020-2024 level and on the Sen's slope.

3.4 Spatial regression

The response is the 2020-2024 upazila mean (and the Sen's slope as a second response). Predictors were standardised so each effect is per one standard deviation: log population density, urban share, distance to Dhaka, log elevation, log major road density, dry-season (December to February) wind speed and annual rainfall. OLS residuals were tested with Moran's I and robust Lagrange multiplier tests, and then SAR, SEM, SLX, SDM and SDEM models were fitted and compared by AIC and likelihood ratio tests.

3.5 Machine learning

Linear regression, random forest (500 trees) and XGBoost were trained on the same covariates. Each was scored twice: with random 10-fold cross-validation, and with spatial 10-fold cross-validation where upazilas are grouped into 10 regions by k-means on their coordinates and one region is held out at a time. Neighbouring upazilas have almost the same air, so random folds let a model learn from the neighbours of the upazila it is predicting. The spatial score is the one to quote. SHAP values explain the XGBoost model.

3.6 Exposure bands and life expectancy

People were counted in nine annual bands from the 0.01° grid (WHO guideline, the four interim targets, then 35-50, 50-75, 75-100 and above 100). The cut needed to reach a target is 1 minus target divided by PM2.5, averaged over people. Life expectancy lost follows the AQLI method (Ebenstein et al. 2017): 0.098 years per μg/m³ above the WHO guideline, averaged over people using the 2020-2024 mean.

CHAPTER 4

How much PM2.5 people breathe

The population-weighted national mean was 39.2 µg/m³ in 1998 and 64.9 µg/m³ in 2024. The 2020-2024 average of 67.6 µg/m³ is 13.5 times the WHO guideline. The area-weighted mean is about 4 to 6 µg/m³ lower in every year, because more people live in the dirtier parts of the country.

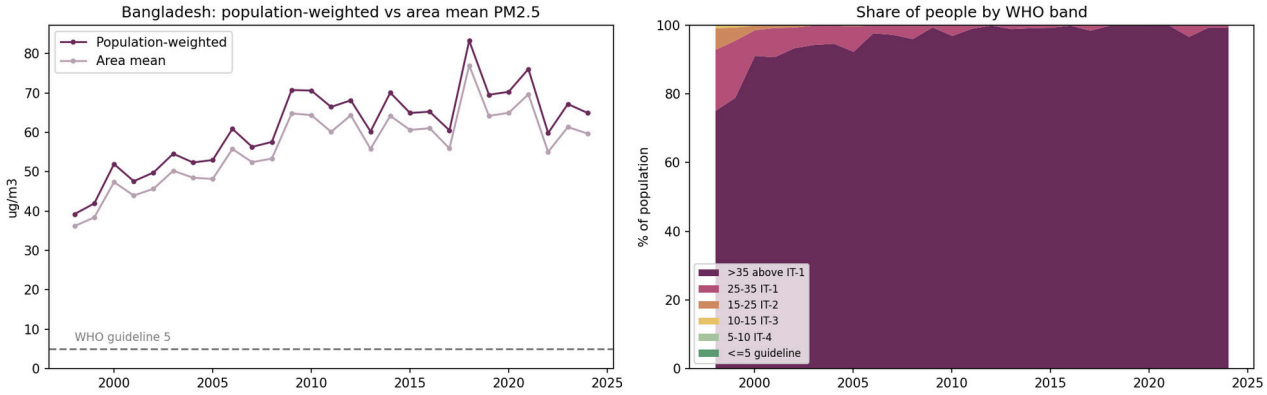


Figure 2. Left: national population-weighted and area-weighted PM2.5, 1998-2024, with the WHO guideline. Right: share of people in each WHO band.

4.1 Divisions

Dhaka division is the most polluted and Sylhet the cleanest by a wide margin. All divisions rose by 17 to 25 µg/m³ between the first and last five years, and no division changed its rank.

Table 5. Divisions, population-weighted PM2.5 (µg/m³).

Division	People (M)	1998-2002	2020-2024	Change	Rank
Dhaka	45.1	54.4	79.8	+25.4	1
Barishal	9.3	50.2	67.1	+16.9	2
Chattogram	33.8	46.8	66.3	+19.5	3
Rajshahi	20.7	43.8	65.9	+22.1	4
Rangpur	17.9	43.5	64.8	+21.3	5
Khulna	17.8	41.5	64.2	+22.7	6
Mymensingh	12.5	41.4	60.2	+18.8	7
Sylhet	11.2	27.6	44.8	+17.2	8
Bangladesh	168.3	46.1	67.6	+21.6	

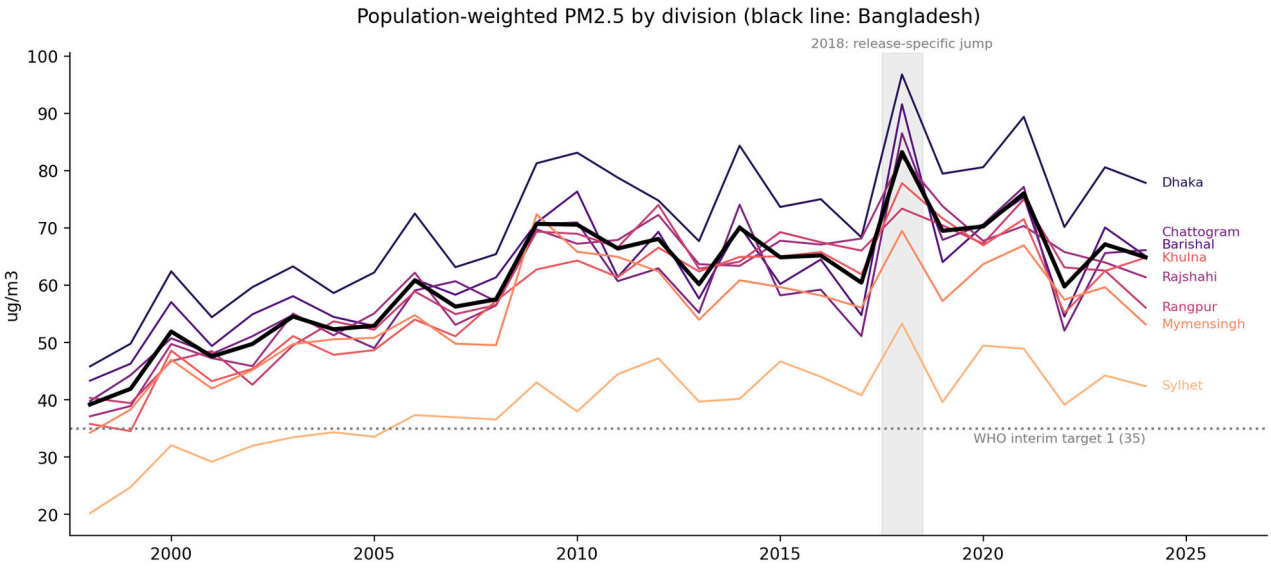


Figure 3. Population-weighted PM2.5 by division, 1998-2024 (black line: Bangladesh). The grey band marks the 2018 release-specific spike.

4.2 Districts and upazilas

The ten most polluted districts are all in Dhaka division except Chandpur. The cleanest are in Sylhet division and the Chittagong Hill Tracts. Sylhet district, at 39.5 µg/m³, is the only district below 40, and it is still eight times the WHO guideline.

Table 6. Most and least polluted districts, 2020-2024 (µg/m³).

#	Most polluted	PM2.5	#	Least polluted	PM2.5
1	Dhaka	83.9	64	Sylhet	39.5
2	Narayanganj	82.5	63	Bandarban	41.0
3	Gazipur	82.3	62	Sunamganj	42.1
4	Manikganj	82.2	61	Rangamati	44.1
5	Munshiganj	81.5	60	Moulvibazar	46.9
6	Narsingdi	79.8	59	Netrakona	50.2
7	Madaripur	78.5	58	Cox's Bazar	51.0
8	Chandpur	77.0	57	Khagrachhari	51.1
9	Faridpur	76.9	56	Habiganj	54.7
10	Shariatpur	76.8	55	Satkhira	55.0

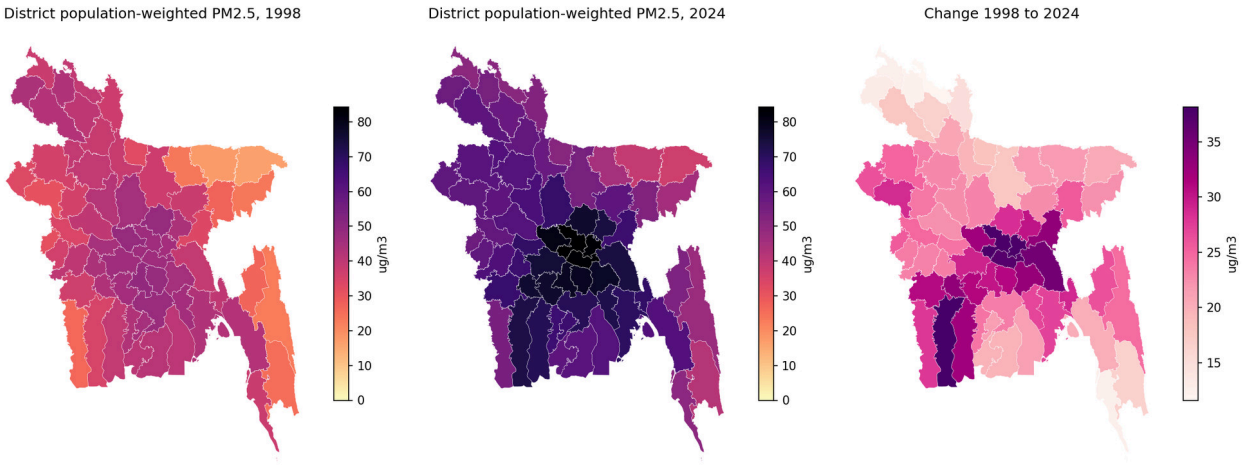


Figure 4. District population-weighted PM2.5 in 1998 and 2024, and the change between them.

At upazila level the worst places are the industrial ring around Dhaka: Savar (85.3 µg/m³), Gazipur City Corporation (84.6), Rupganj (84.5) and Kaliganj (84.4). The two Dhaka city corporations rose the most among the top upazilas, by about 34 µg/m³ since 1998-2002.

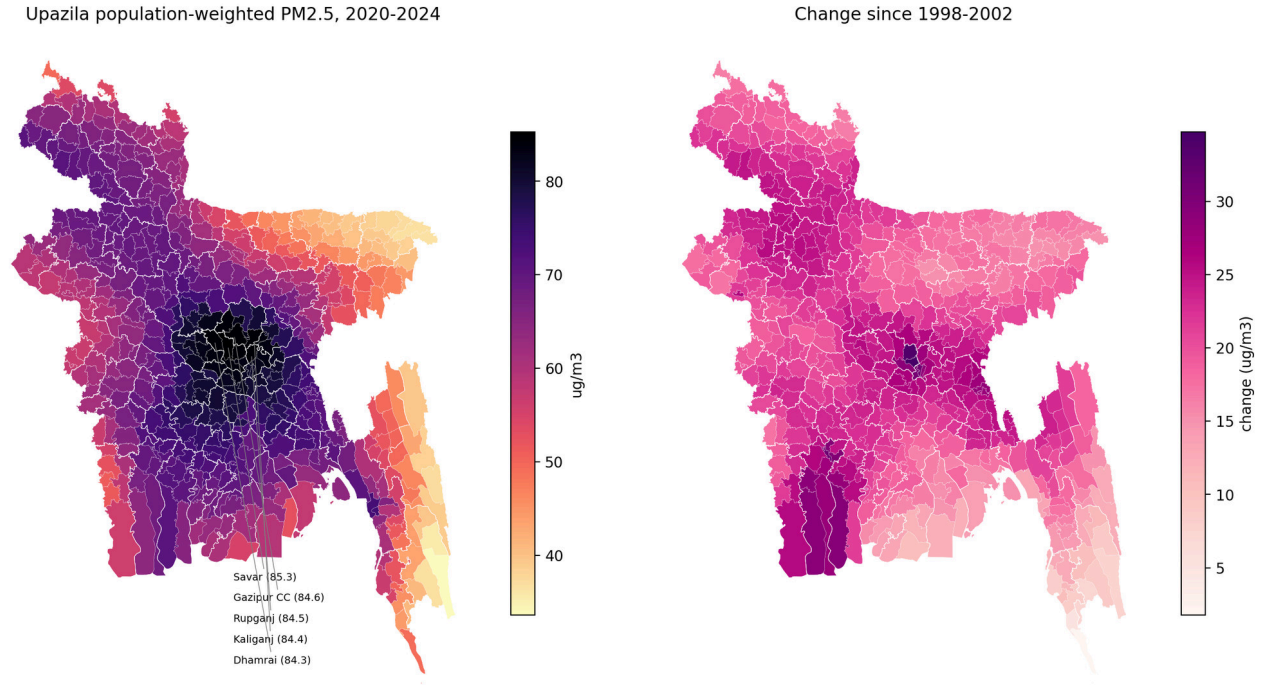


Figure 5. Upazila population-weighted PM2.5 in 2020-2024 (left) and the change since 1998-2002 (right).

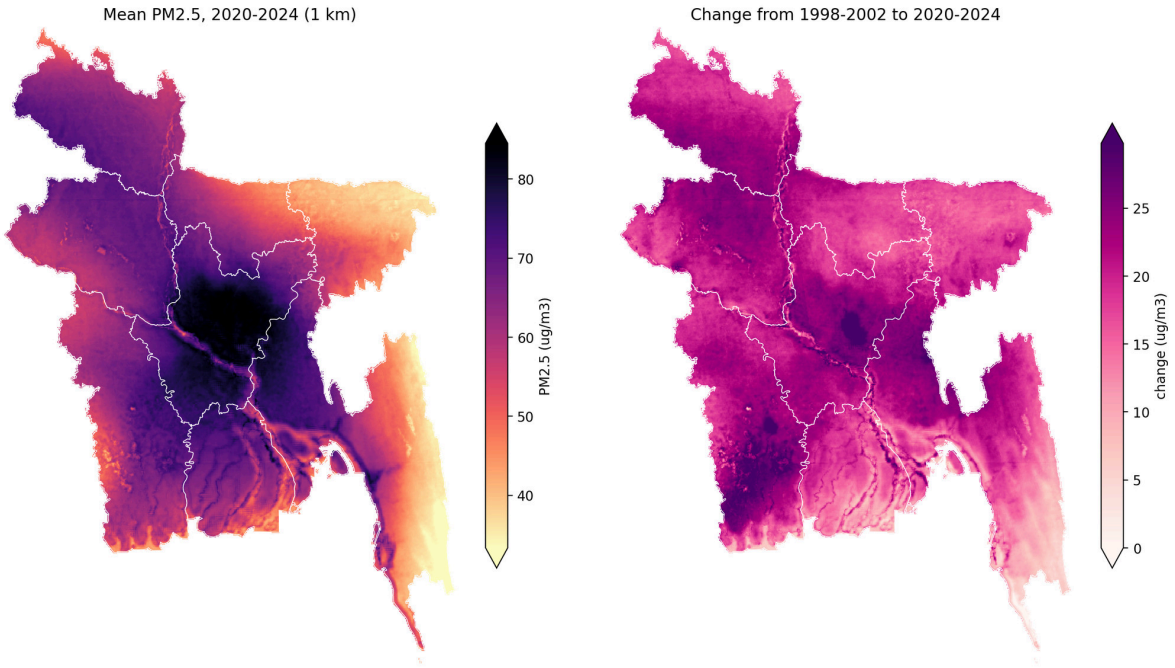


Figure 6. The 1 km PM2.5 surface, 2020-2024 mean, and the change from 1998-2002.

4.3 Where people live matters

The weighting gap is the population-weighted minus the area-weighted mean. I expected it to be largest in the densest districts. It is not (Spearman correlation with density 0.02). It is largest in Bagerhat (+5.5), Khulna (+4.9), Bhola (+4.0), Bandarban (+3.9) and Lakshmipur (+3.7), where large areas of forest, river or char land are empty and cleaner, so the people are concentrated in the dirtier part of the district.

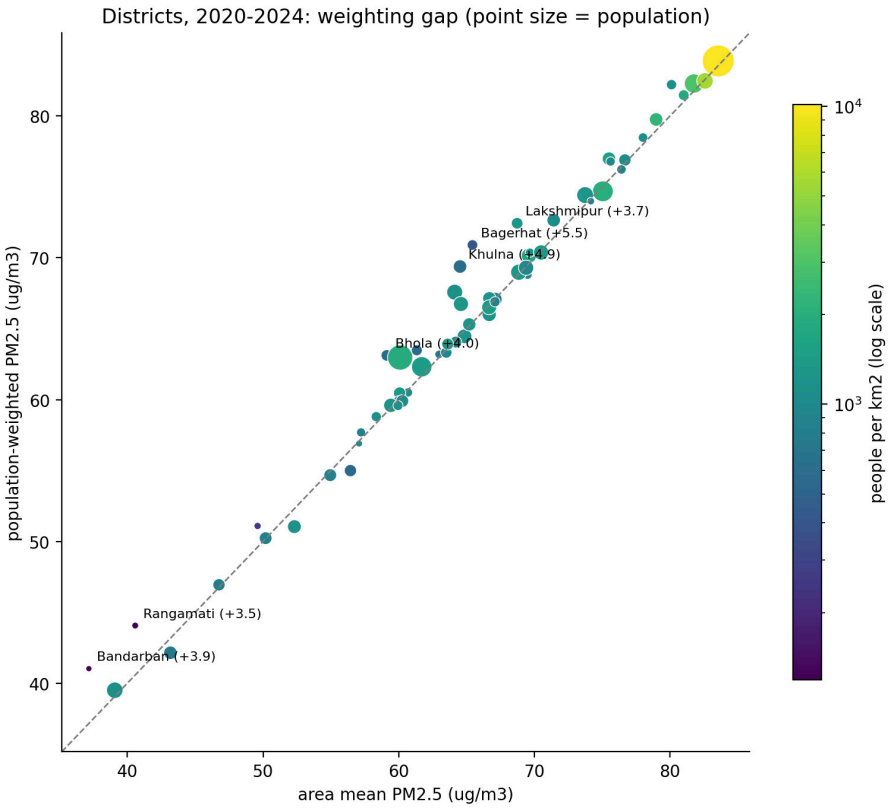


Figure 7. Districts, 2020-2024: population-weighted against area-weighted PM2.5. Points above the line have people in the dirtier part of the district.

CHAPTER 5

People above the WHO targets

Nobody in Bangladesh met the WHO guideline of 5 µg/m³ in any year, and nobody met the 10 µg/m³ target either. The share of people above 35 µg/m³ went from 75% in 1998 to 99.4% in 2024. In 12 of the 27 years at least 99% of people lived above 35, starting in 2009.

Table 7. People by WHO band, selected years (millions).

Year	15-25 (IT-2)	25-35 (IT-1)	Above 35	Share above 35
1998	10.7	29.9	126.5	75.1%
2002	1.0	10.1	157.2	93.4%
2006	0.0	3.8	164.6	97.7%
2010	0.0	5.1	163.3	97.0%
2014	0.0	1.3	167.1	99.2%
2018	0.0	0.1	168.3	100.0%
2022	0.0	5.6	162.8	96.7%
2024	0.0	1.0	167.3	99.4%

1998 also had 1.25 million people in the 10-15 band. No one was below 10 in any year.

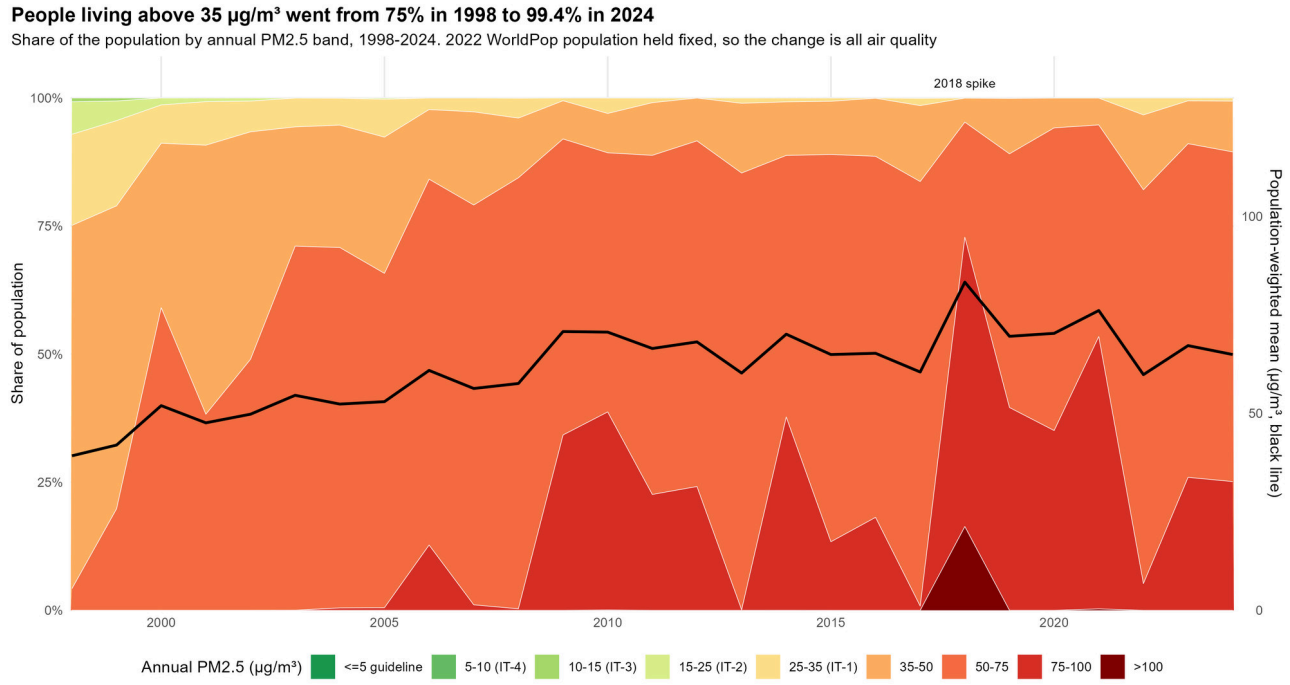


Figure 8. Share of the population by annual PM2.5 band, 1998-2024, with the population-weighted mean (black line).

5.1 How the distribution moved

The median person breathed 40 µg/m³ in 1998 and 63 µg/m³ in 2024. The whole distribution shifted right. The cleanest-air 10% of people were at 49 µg/m³ or below in 2024 and the dirtiest 10% at 82 or above. 2010 had a heavier upper tail than 2024 (P90 90 against 82), which matches the plateau in Chapter 6.

Table 8. Exposure percentiles across people (µg/m³).

Year	P10	P25	Median	P75	P90
1998	27.6	35.0	40.5	45.2	48.4
2010	48.4	62.8	72.1	80.9	90.0
2024	49.5	57.3	63.1	75.1	81.9

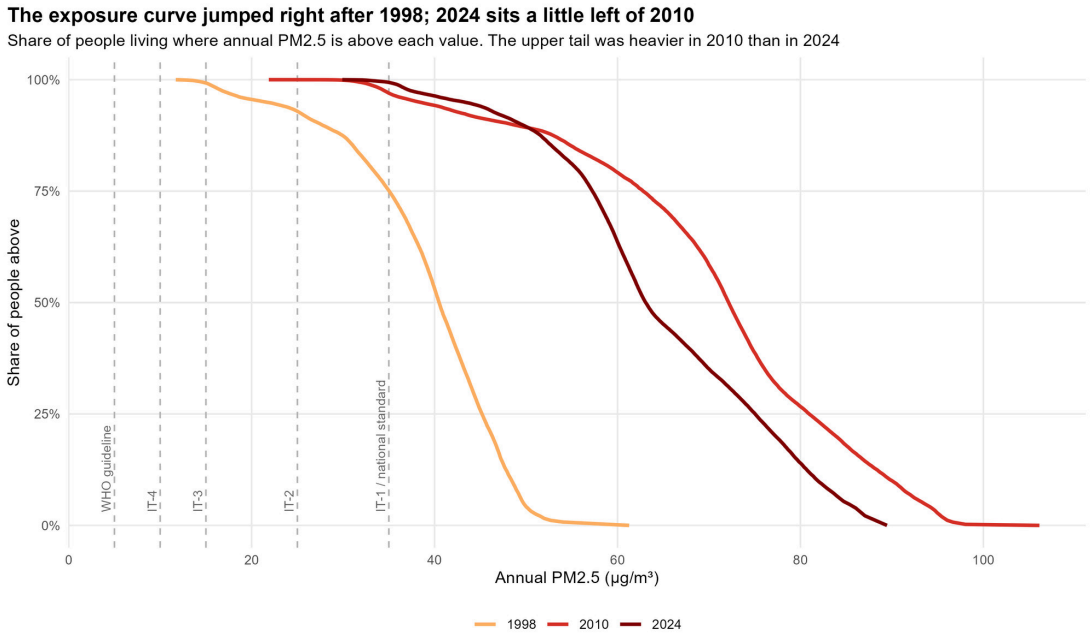


Figure 9. Share of people living where annual PM2.5 is above each value, 1998, 2010 and 2024.

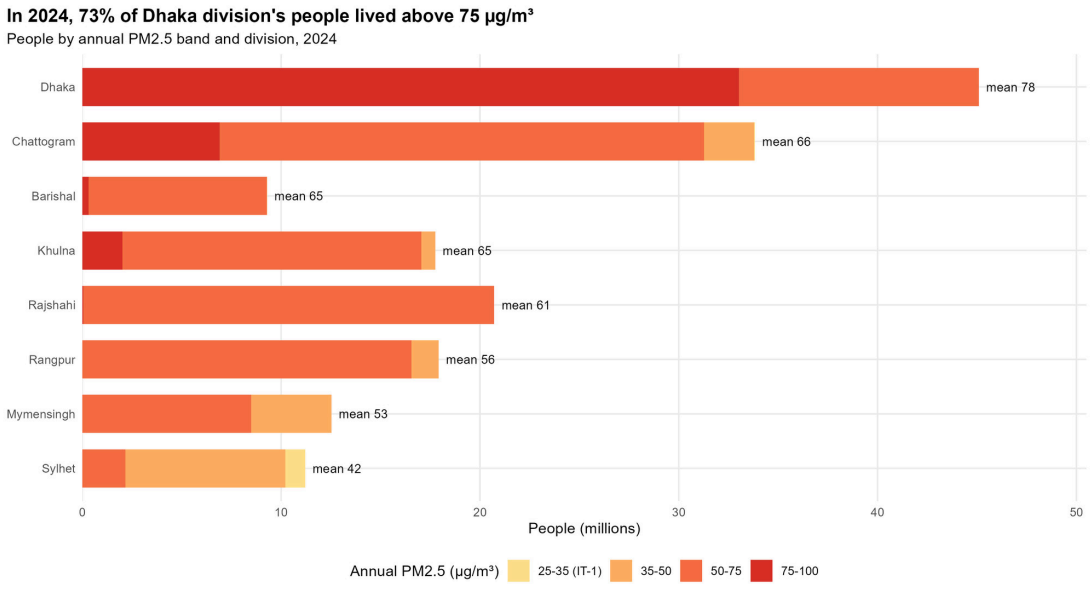


Figure 10. People by band and division, 2024. 73% of Dhaka division's people lived above 75 µg/m³.

CHAPTER 6

Trends since 1998

National PM2.5 rose significantly over 1998-2024 (Mann-Kendall tau 0.56, corrected Z 4.30, p 1.7e-05). The Sen's slope is +1.08 µg/m³ per year (95% CI 0.65 to 1.41), or 23% of the 1998-2002 level per decade. Dropping 2018 changes it to 1.07.

6.1 A rise and then a plateau

The trend lines in Figure 11 are not straight. Splitting the period shows why. Every division rose fast until 2010 (Sen's slope 1.3 to 2.5 µg/m³ per year, all p < 0.001), and none has a significant trend after 2010. Nationally the slope fell from 2.15 to 0.06 µg/m³ per year. So exposure climbed to a high level and stayed there, it did not keep rising. The STL trend component (Figure 18) shows the same shape.

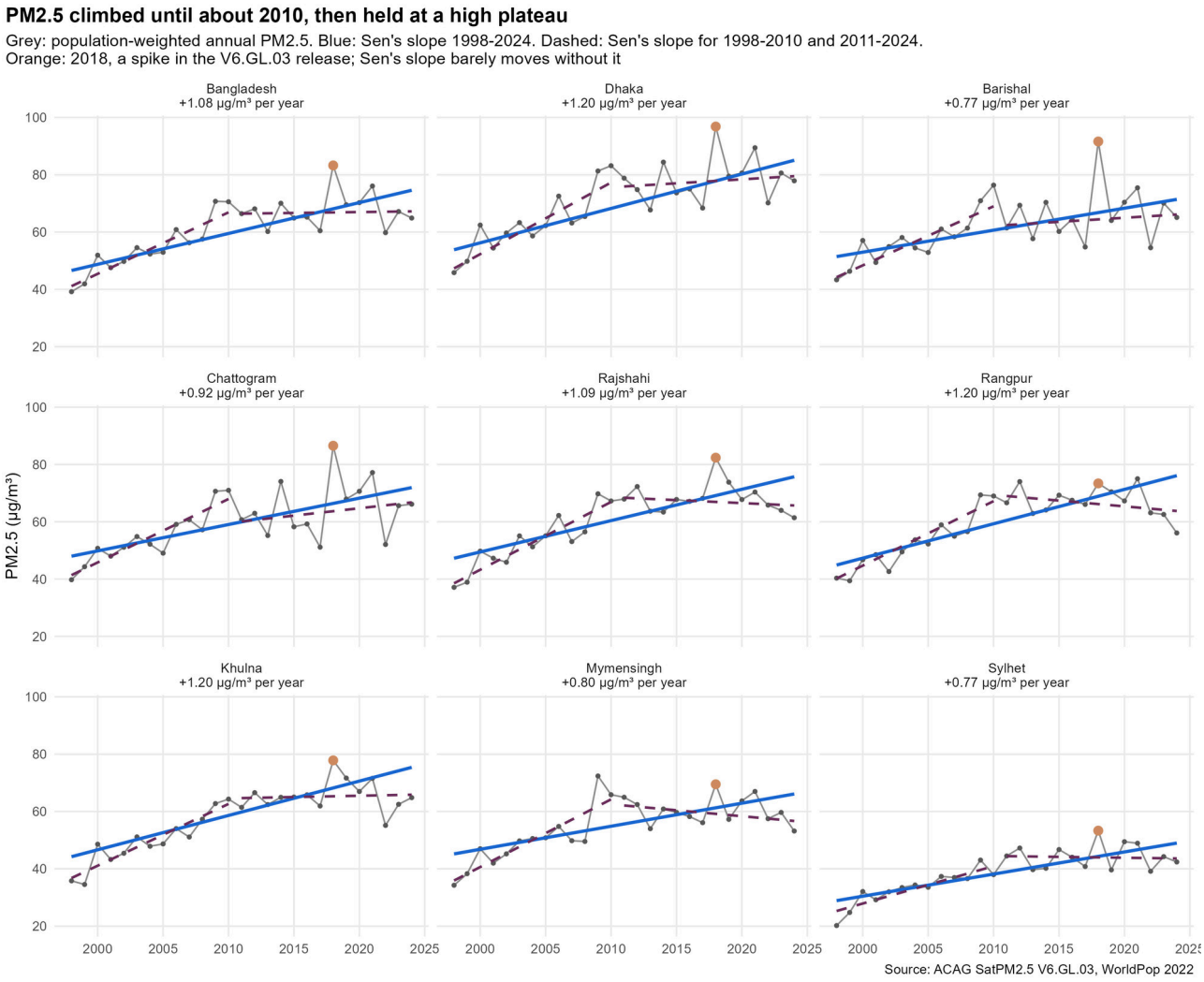


Figure 11. Annual PM2.5 by division with Sen's slope for 1998-2024 (blue) and separate slopes for 1998-2010 and 2011-2024 (dashed). Orange: 2018.

Table 9. Sen's slope by division (µg/m³ per year).

Area	1998-2024	95% CI	% per decade	1998-2010	2011-2024	p 2011-2024
Bangladesh	+1.08	0.65 to 1.41	23	+2.15	+0.06	0.83
Dhaka	+1.20	0.75 to 1.62	22	+2.50	+0.28	0.44
Rangpur	+1.20	0.66 to 1.46	28	+2.25	-0.40	0.38
Khulna	+1.20	0.88 to 1.57	29	+2.16	+0.09	0.66
Rajshahi	+1.09	0.67 to 1.52	25	+2.37	-0.21	0.44
Chattogram	+0.92	0.54 to 1.39	20	+2.21	+0.51	0.51
Mymensingh	+0.80	0.46 to 1.18	19	+2.36	-0.42	0.38
Sylhet	+0.77	0.53 to 1.07	28	+1.30	-0.06	0.83
Barishal	+0.77	0.37 to 1.23	15	+2.06	+0.28	0.58

6.2 Where it rose fastest

500 of the 507 upazilas have a significant rise after the Benjamini-Hochberg correction. The fastest absolute rises are in the north-west and on the Khulna coast, with Khulna City Corporation at +1.59 µg/m³ per year. The slowest is Teknaf at +0.17. In relative terms the picture changes: Sylhet and the north-east started low and rose fast in percentage terms.

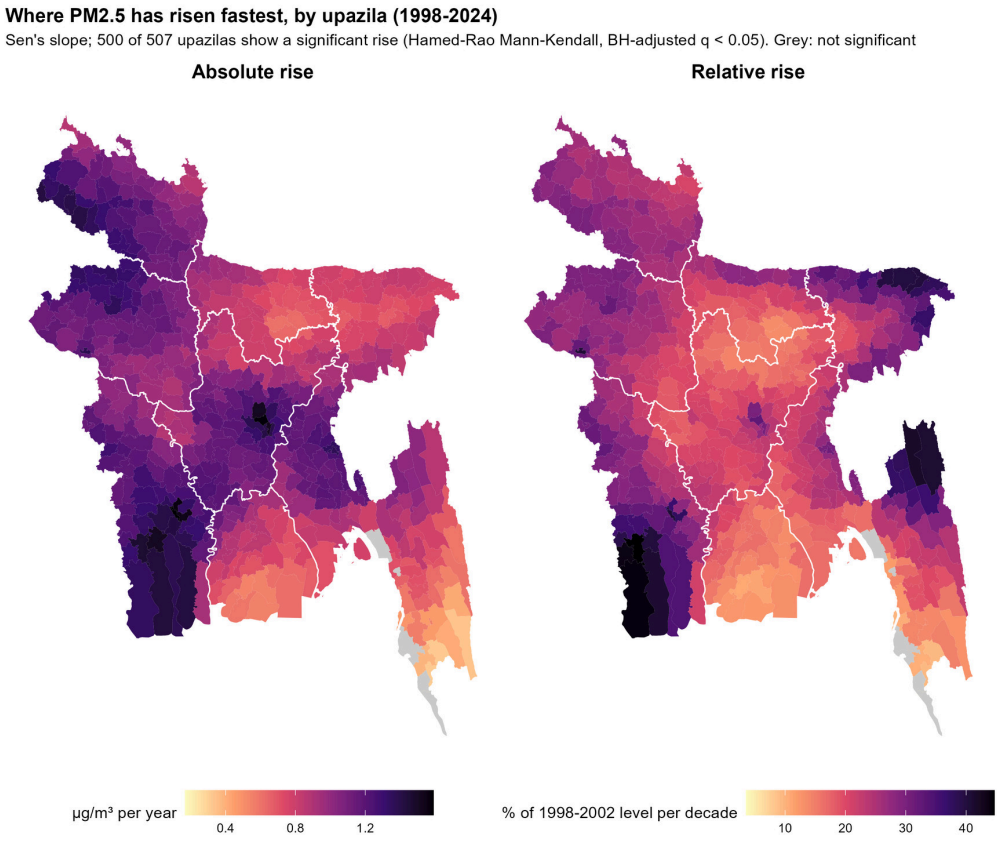


Figure 12. Sen's slope by upazila, 1998-2024, absolute (left) and relative to the 1998-2002 level (right). Grey: not significant.

Upazilas that started dirtier rose slightly faster in absolute terms (Spearman rho 0.26) but slower in relative terms (rho -0.55). So the gap between clean and dirty places did not close in μg/m³, but it narrowed in percentage terms.

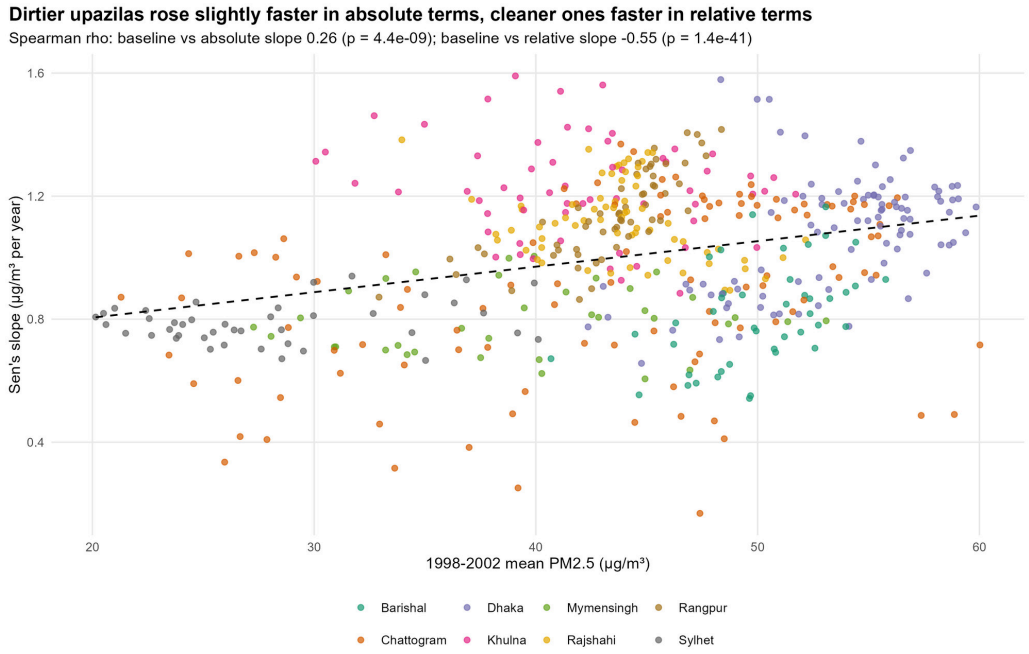


Figure 13. 1998-2002 level against Sen's slope for every upazila.

CHAPTER 7

Seasonality

The season matters more than the year. Nationally, January averages 121 µg/m³ and July 26 µg/m³, so the worst month is about 4.7 times the cleanest. The dry season (November to February) is when brick kilns run, rain is low and north-westerly winds bring air from the Indo-Gangetic Plain. The monsoon washes the air out.

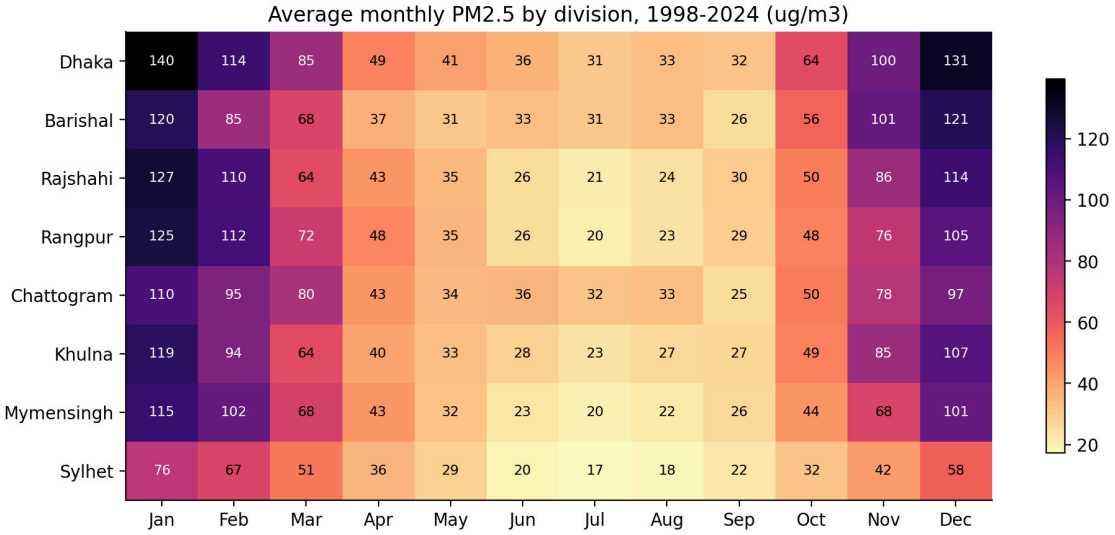


Figure 14. Average monthly PM2.5 by division, 1998-2024 (µg/m³).

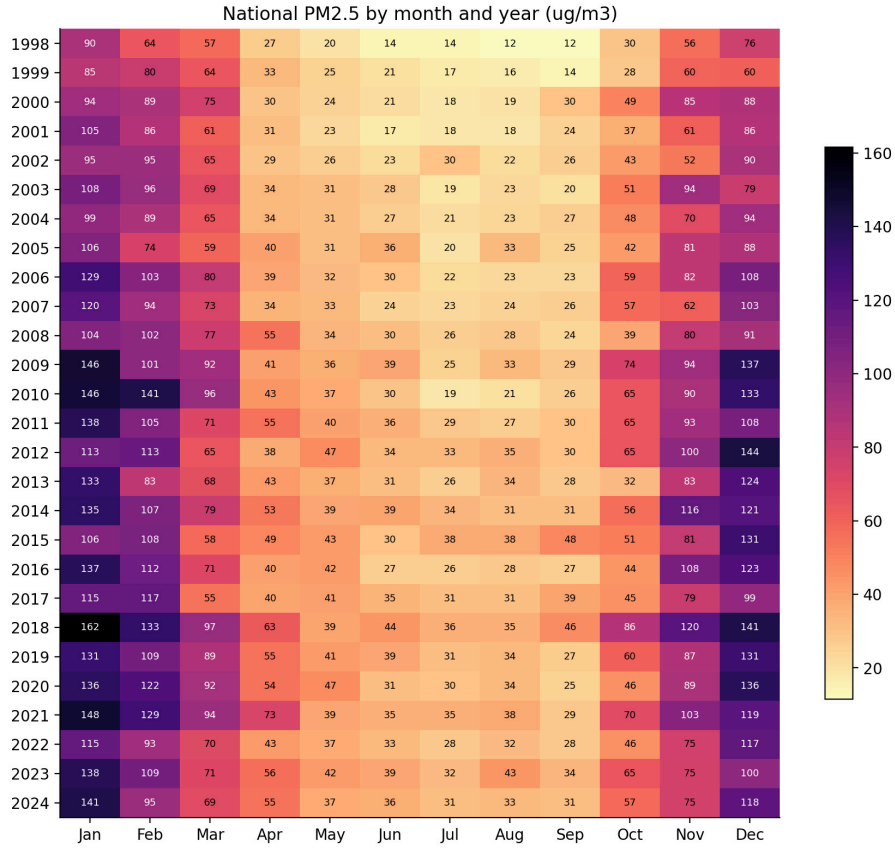


Figure 15. National PM2.5 by month and year (µg/m³).

7.1 Which months rose

December rose the most in absolute terms (+2.15 µg/m³ per year), followed by January (+1.91). In relative terms the monsoon months rose most: August by 46% of its 1998-2002 level per decade. The seasonal Mann-Kendall test is significant in every division.

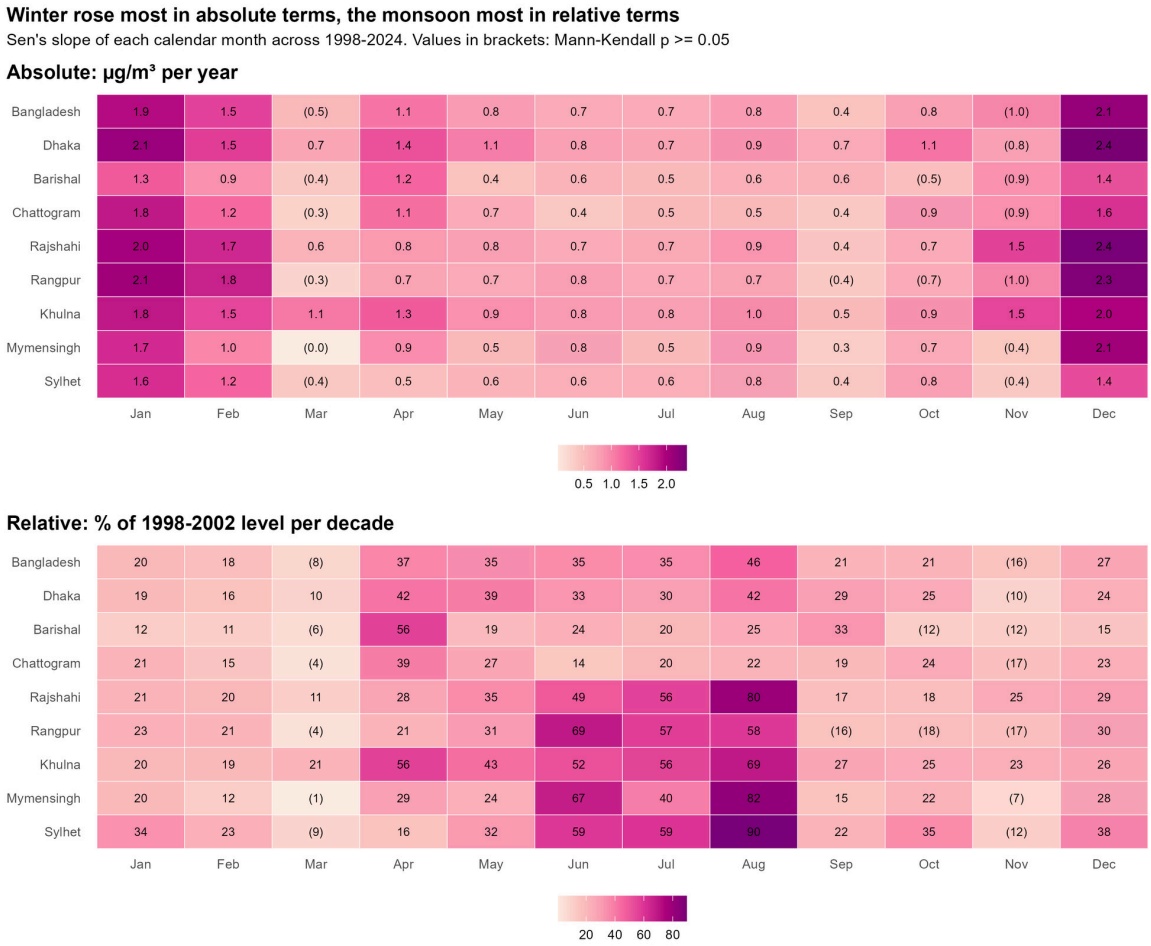


Figure 16. Sen's slope by calendar month and division: absolute (top) and relative to the 1998-2002 level (bottom). Values in brackets are not significant.

7.2 The seasonal swing is narrowing

STL decomposition shows a strong seasonal component (strength 0.94) and a weaker trend (0.65). The ratio between the peak and trough months fell from 5.6 in 1998 to 4.6 in 2024 nationally, and it narrowed in every division except Chattogram. The biggest drops were in Rajshahi (8.6 to 5.5) and Rangpur (8.0 to 5.3). Monsoon air got dirtier faster than winter air.

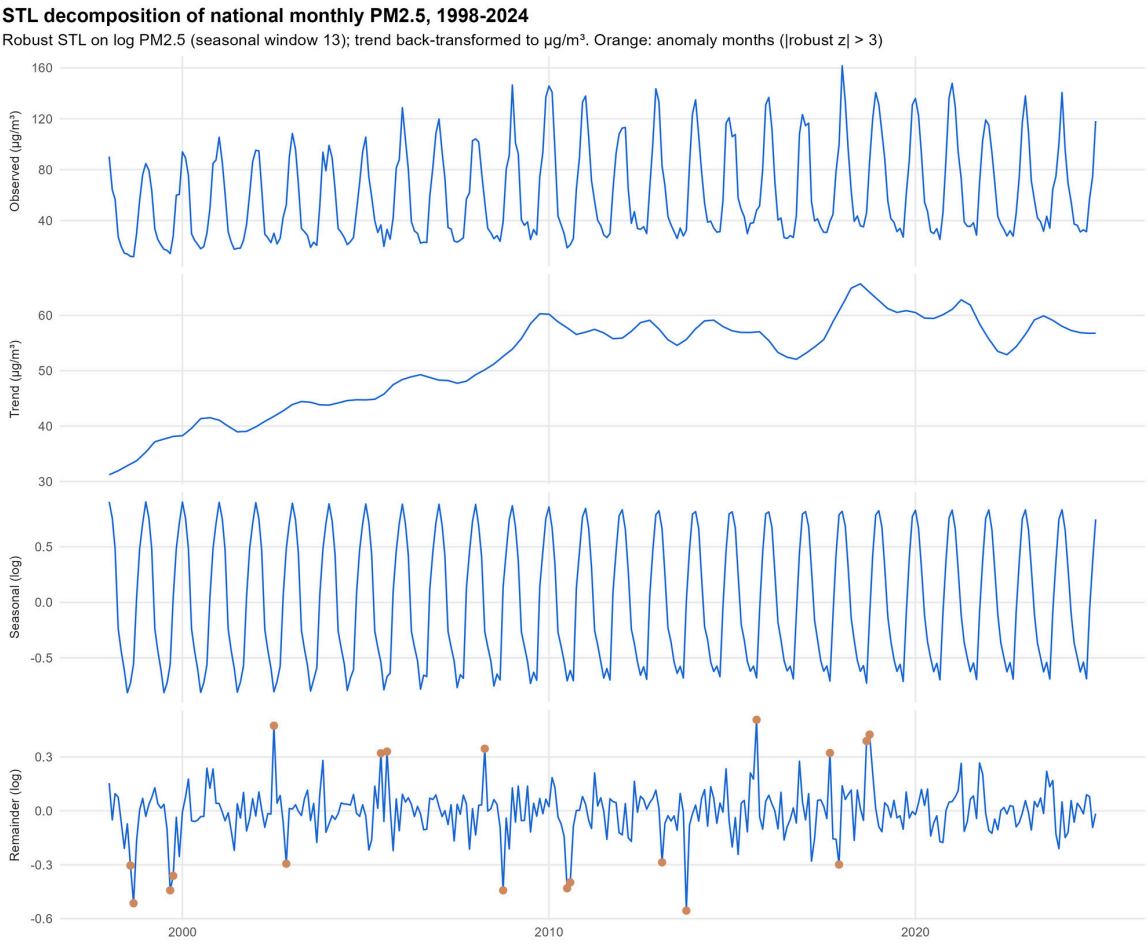


Figure 17. STL decomposition of national monthly PM2.5: observed, trend, seasonal and remainder. Orange: anomaly months (robust z above 3).

Table 10. STL summary by division.

Area	Trend 1998	Trend 2024	Swing 1998	Swing 2024	Peak	Trough
Bangladesh	32.8	57.1	5.6x	4.6x	Jan	Sep
Dhaka	37.5	68.7	5.4x	4.3x	Jan	Jul
Barishal	35.7	57.5	5.6x	4.5x	Dec	Sep
Chattogram	35.5	59.2	4.2x	4.6x	Jan	Sep
Rajshahi	29.3	52.2	8.6x	5.5x	Jan	Jul
Rangpur	31.1	47.9	8.0x	5.3x	Jan	Jul
Khulna	27.7	56.3	7.3x	4.6x	Jan	Jul
Mymensingh	26.6	43.6	6.6x	5.2x	Jan	Jul
Sylhet	18.4	37.1	6.5x	4.0x	Jan	Jul

Trend: STL trend component, µg/m³. Swing: peak month over trough month of the seasonal component.

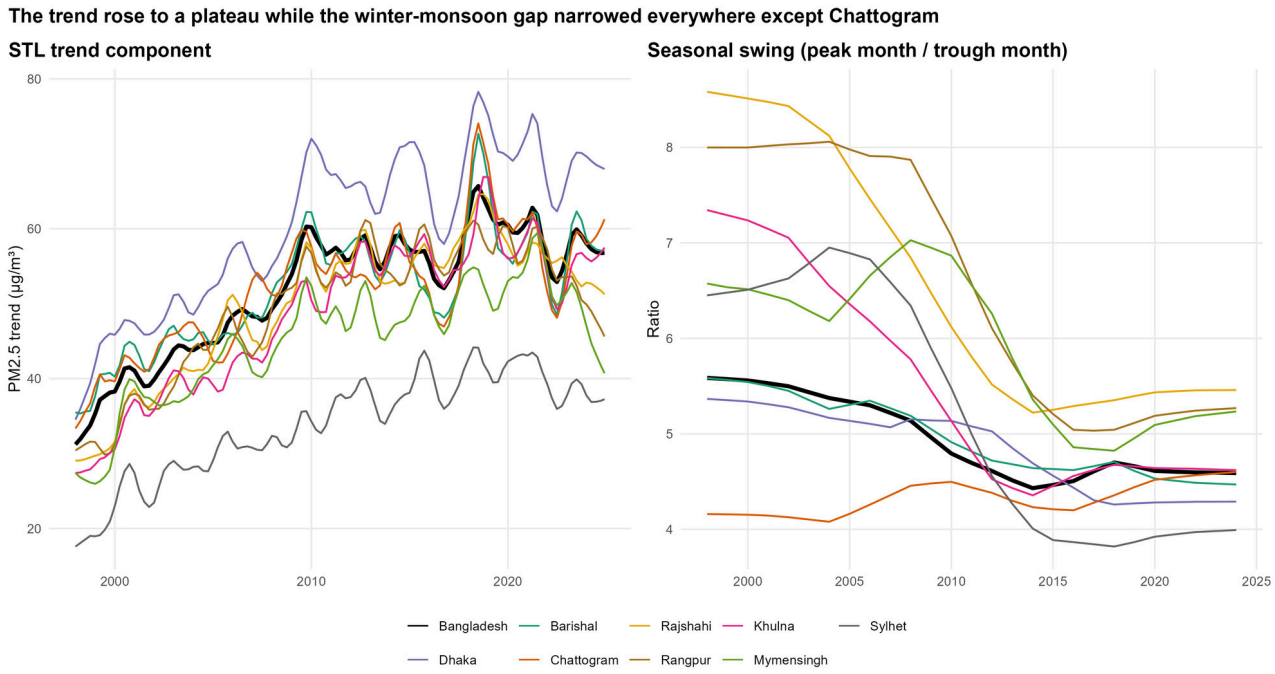


Figure 18. STL trend component (left) and seasonal swing (right) by division.

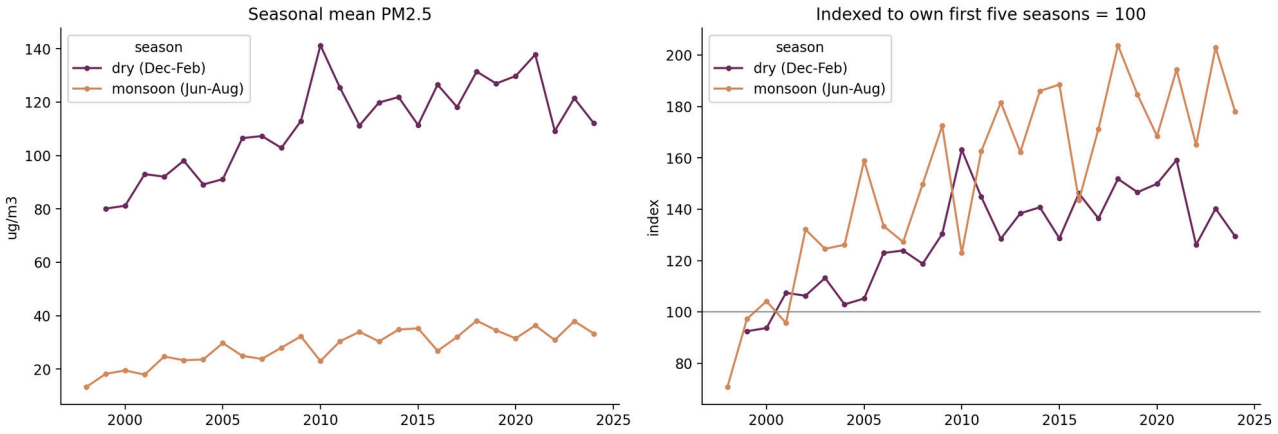


Figure 19. Seasonal means (left) and the same series indexed to each season's first five years (right). Dry: December-February. Monsoon: June-August.

Indexed to their own first five seasons, monsoon PM2.5 ended near 180 and the dry season near 140. Monthly values come from the coarser 0.1° grid. For a few small coastal upazilas the average of the months differs from the annual 0.01° value by up to 11 µg/m³, so seasonal results are reported for divisions and districts only.

CHAPTER 8

Spatial clustering

Upazilas look very much like their neighbours. Global Moran's I is 0.95 for the 2020-2024 level and 0.91 for the Sen's slope with queen weights (0.92 and 0.89 with 6 nearest neighbours, permutation p 0.001 for all). Part of this very high value comes from the satellite surface itself, which is smooth at the upazila scale, so I read it as strong regional structure rather than 507 independent readings.

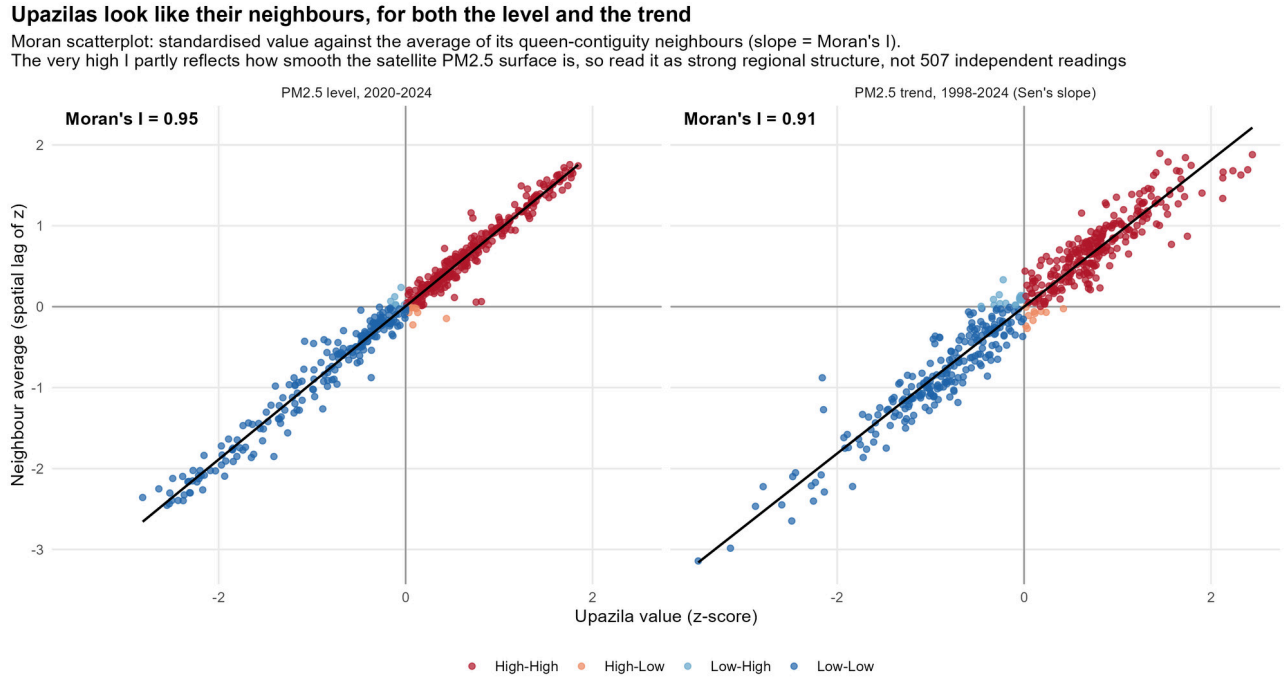


Figure 20. Moran scatterplots for the level and the trend. The slope of the line is Moran's I.

8.1 How far it reaches

The correlogram falls steadily with distance. Moran's I drops below 0.1 at neighbour order 7, about 124 km, for both the level and the trend. Beyond about 140 km the level becomes negatively correlated, which reflects the contrast between the Dhaka basin and the cleaner north-east and south-east.

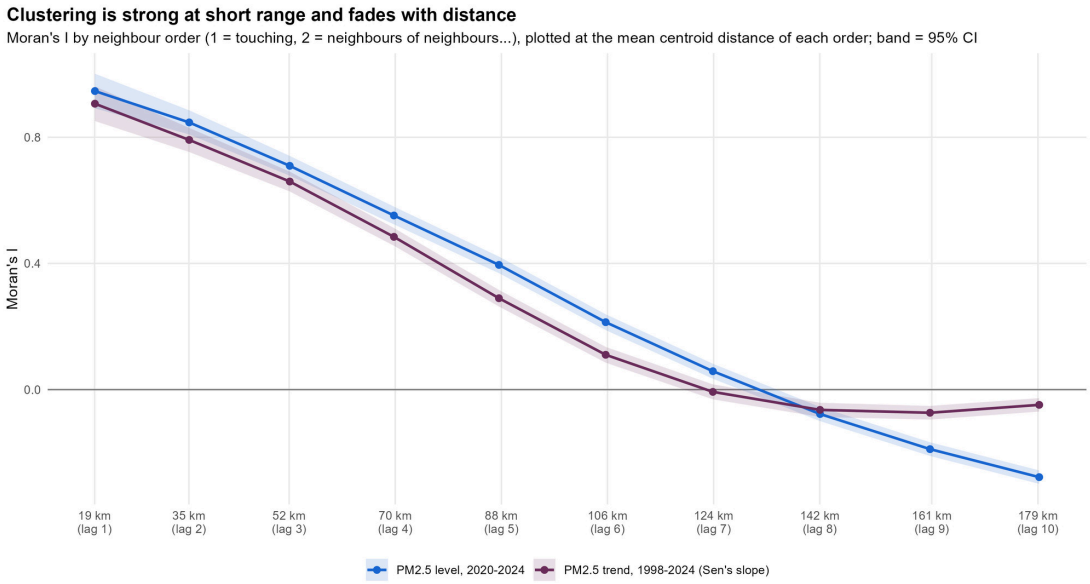


Figure 21. Moran's I by neighbour order, plotted at the mean centroid distance of each order (95% CI band).

8.2 Hot and cold clusters

Table 11. LISA clusters ($p < 0.05$).

Variable	Cluster	Upazilas	People (M)	Share of people	Mean PM2.5	Mean slope	Main division
Level	High-High	117	53.9	32%	79.8	+1.20	Dhaka
Level	Low-Low	88	20.7	12%	47.3	+0.73	Sylhet
Level	Not significant	302	93.8	56%	65.2	+1.02	
Trend	High-High	133	55.0	33%	75.1	+1.29	Khulna
Trend	Low-Low	119	35.4	21%	56.9	+0.71	Chattogram
Trend	Not significant	255	77.9	46%	67.3	+1.02	

The level has one large hot cluster centred on Dhaka and two cold clusters, Sylhet in the north-east and the Chattogram coast and Hill Tracts in the south-east. The trend has a different pattern: hot clusters in the north-west (Rajshahi and Rangpur) and in the south-west around Khulna. So the places with the dirtiest air today are not all the places where it is getting worse fastest. Gi* finds 77 upazilas in hot spots at 99% confidence for the level and 62 for the trend.

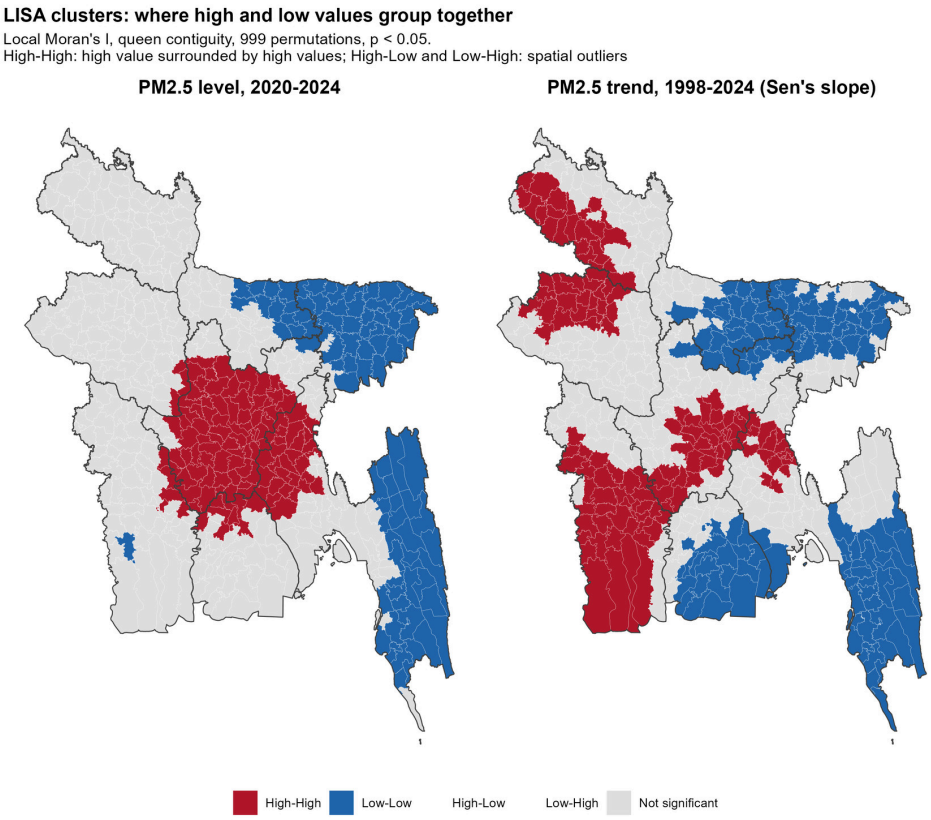


Figure 22. LISA clusters for the 2020-2024 level and the 1998-2024 trend.

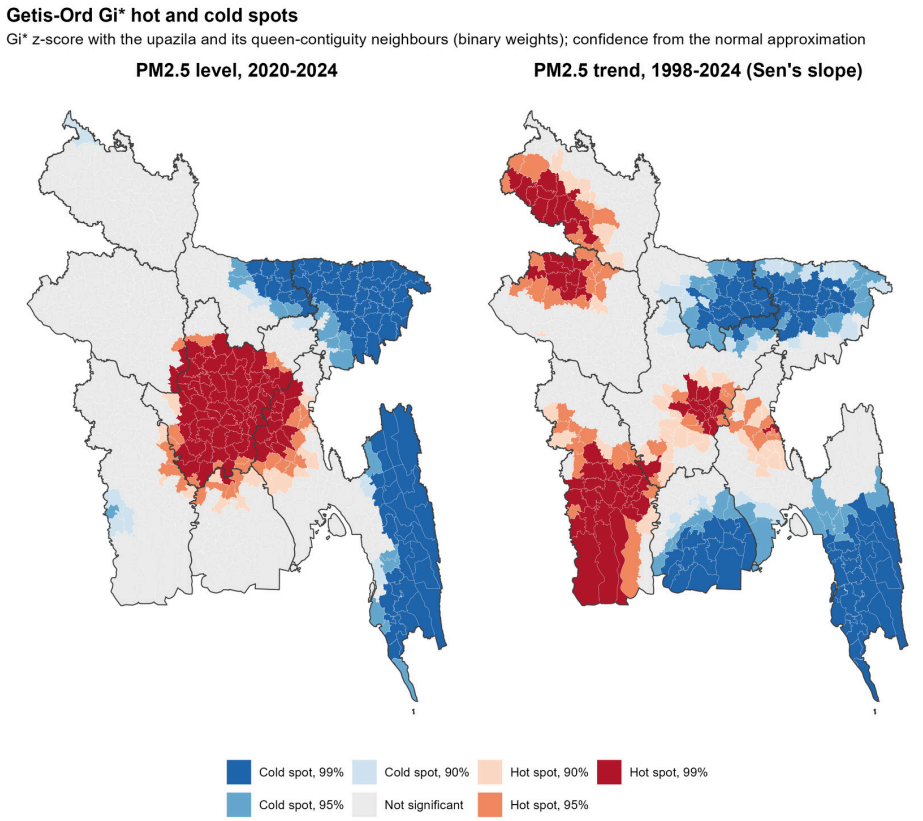


Figure 23. Getis-Ord Gi* hot and cold spots for the level and the trend.

CHAPTER 9

Spatial regression

OLS on the seven standardised covariates explains about 65% of the variation in the 2020-2024 level, but its residuals are strongly clustered (Moran's I 0.84). Both robust LM tests are significant (lag 43.4, p 4.5e-11; error 17.0, p 3.8e-05), so a spatial model is needed.

Table 12. Model comparison, 2020-2024 level.

Model	Parameters	AIC	ΔAIC	rho / lambda	Pseudo R^2	RMSE	Residual Moran's I
OLS	9	3365.4	1369.4		0.65	6.57	0.84
SLX	16	3274.3	1278.3		0.71	5.92	0.83
SAR	10	2097.7	101.7	rho 0.97	0.98	1.57	0.11
SEM	10	2012.6	16.6	lambda 0.99	0.98	1.42	0.07
SDM	17	1996.0	0	rho 0.97	0.98	1.40	0.06
SDEM	17	2002.5	6.5	lambda 0.99	0.98	1.39	0.07

Pseudo R^2 includes the spatial term, so it is high for every spatial model and is not a measure of what the covariates explain.

9.1 Which model to read

The spatial Durbin model has the lowest AIC. But its rho of 0.97 means a spatial multiplier of about 38, and the direct and indirect impacts from such a model are unstable and hard to interpret. The spatial error model is only 16.6 AIC points behind, removes almost all of the residual clustering (Moran's I 0.07), and its coefficients read like ordinary regression effects. I interpret the SEM and report the SDM as a check.

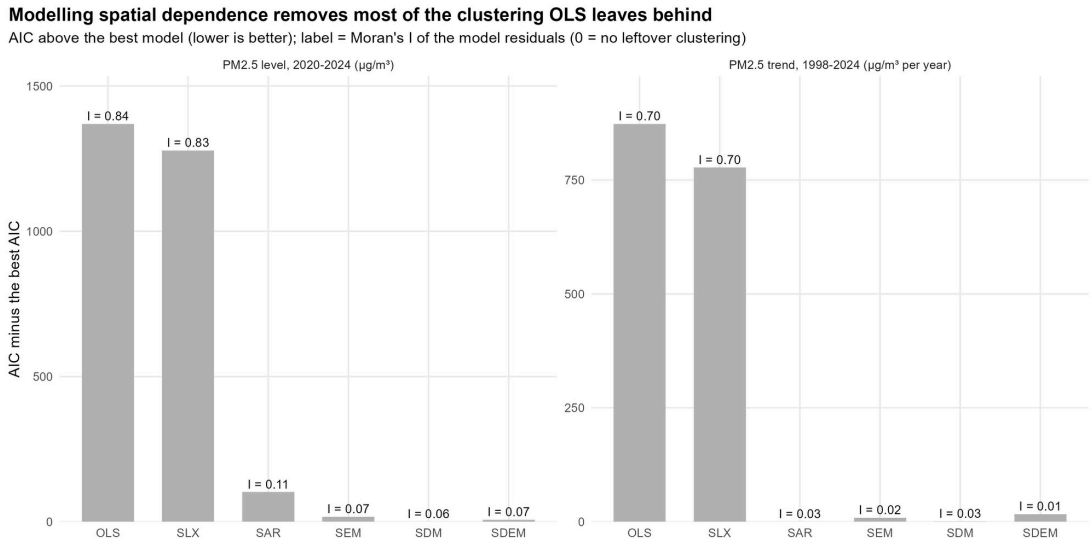


Figure 24. AIC above the best model and the Moran's I of each model's residuals, for the level and the trend.

9.2 Effects

Table 13. Effect per 1 SD of each predictor on the 2020-2024 level (µg/m³).

Predictor	1 SD	OLS	SEM	SEM 95% CI	SEM p
Distance to Dhaka	74 km	-5.39	-7.88	-9.48 to -6.28	<0.001
Elevation (log)	0.87 log m	+0.08	-1.33	-1.91 to -0.74	<0.001
Rainfall	2.1 mm/day	-4.84	-1.06	-1.66 to -0.46	0.001
Population density (log)	0.34	+2.36	-0.41	-0.80 to -0.01	0.045
Dry-season wind speed	0.42 m/s	+3.71	+0.37	+0.03 to +0.72	0.034
Urban share	0.27	-0.96	+0.29	-0.03 to +0.61	0.071
Major road density (log)	0.15	+0.60	+0.25	+0.05 to +0.44	0.013

Distance to Dhaka is the one strong and stable effect: it gets larger once spatial dependence is modelled. Rainfall and dry-season wind look important in OLS but shrink by about 80 to 90% in the SEM, because they vary smoothly across the country and OLS was using them to stand in for regional structure. Elevation becomes clearly negative (higher ground in the Hill Tracts and the north-east is cleaner). Density and urban share add little at upazila level. These are associations between places, not causal effects.

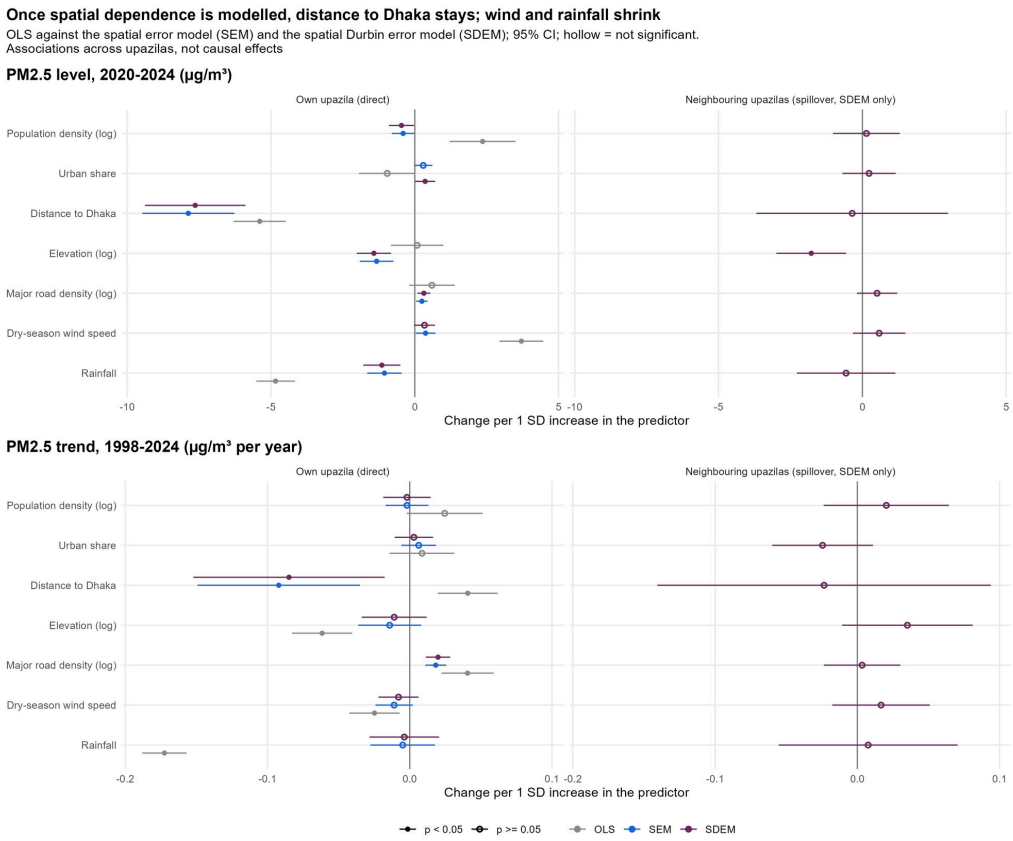


Figure 25. Effects per 1 SD in OLS, SEM and SDEM (own upazila and spillover from neighbours), for the level (top) and the trend (bottom).

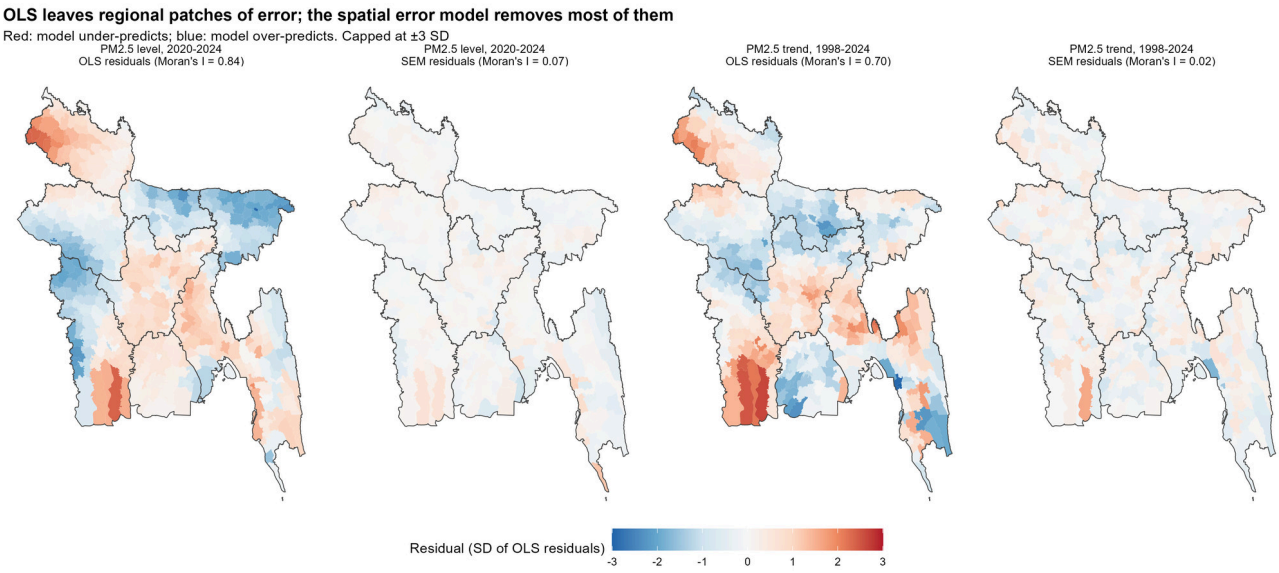


Figure 26. OLS and SEM residuals for the level and the trend. OLS leaves regional patches of error, SEM removes most of them.

For the Sen's slope as the response the SAR has the lowest AIC, and the SEM again removes the residual clustering (0.70 to 0.02). Distance to Dhaka is the only sizeable effect (-0.09 µg/m³ per year per SD).

CHAPTER 10

Machine learning drivers

The same seven covariates were given to three models. Distance to Dhaka has the strongest single correlation with the 2020-2024 level (Spearman -0.67), followed by elevation (-0.39), population density (+0.41) and rainfall (-0.31).

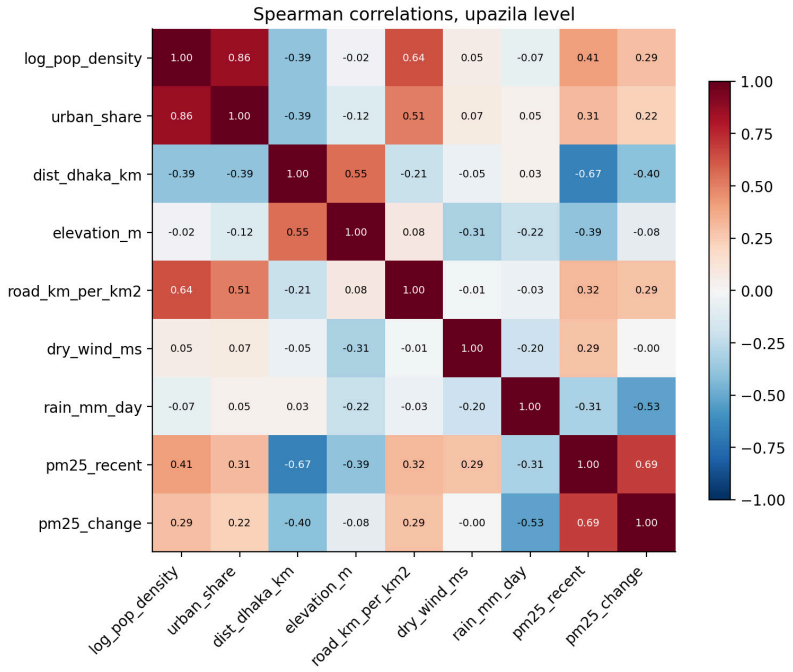


Figure 27. Spearman correlations between covariates and PM2.5 at upazila level.

10.1 Random against spatial cross-validation

Table 14. Out-of-fold scores, 10-fold cross-validation.

Target	Model	R^2 random	R^2 spatial	MAE random	MAE spatial	RMSE spatial
Level 2020-2024	Linear regression	0.64	0.28	5.56	7.81	9.40
Level 2020-2024	Random forest	0.93	0.62	2.29	5.24	6.83
Level 2020-2024	XGBoost	0.94	0.66	2.03	4.71	6.46
Change since 1998-2002	Linear regression	0.59	0.36	2.18	2.78	3.38
Change since 1998-2002	Random forest	0.82	0.35	1.35	2.59	3.40
Change since 1998-2002	XGBoost	0.83	0.40	1.31	2.49	3.25

With random folds XGBoost looks almost perfect (R^2 0.94). When a whole region is held out it drops to 0.66, and for the change since 1998-2002 it drops from 0.83 to 0.40. Most of the apparent skill under random folds came from neighbours looking alike. The residuals confirm it: XGBoost residuals have Moran's I 0.21 under random CV and 0.82 under spatial CV. The spatial scores are the ones that describe how well the covariates predict an unseen area.

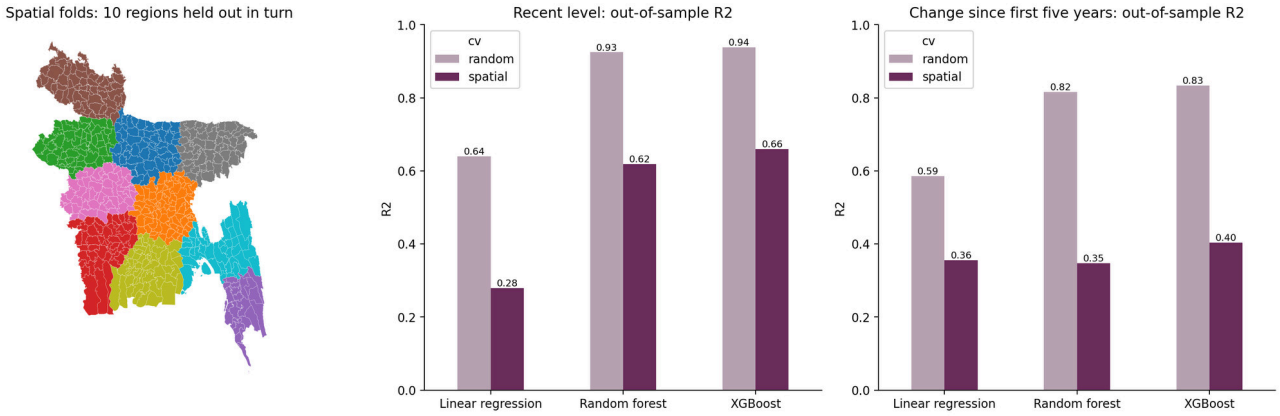


Figure 28. The 10 spatial folds (left) and out-of-fold R^2 under random and spatial cross-validation, for the level and the change.

10.2 What the model leans on

Table 15. Mean absolute SHAP, XGBoost, 2020-2024 level (µg/m³).

Rank	Predictor	Mean |SHAP|
1	Distance to Dhaka	3.94
2	Rainfall	3.11
3	Dry-season wind speed	3.03
4	Elevation	1.48
5	Population density (log)	0.47
6	Urban share	0.46
7	Major road density	0.37

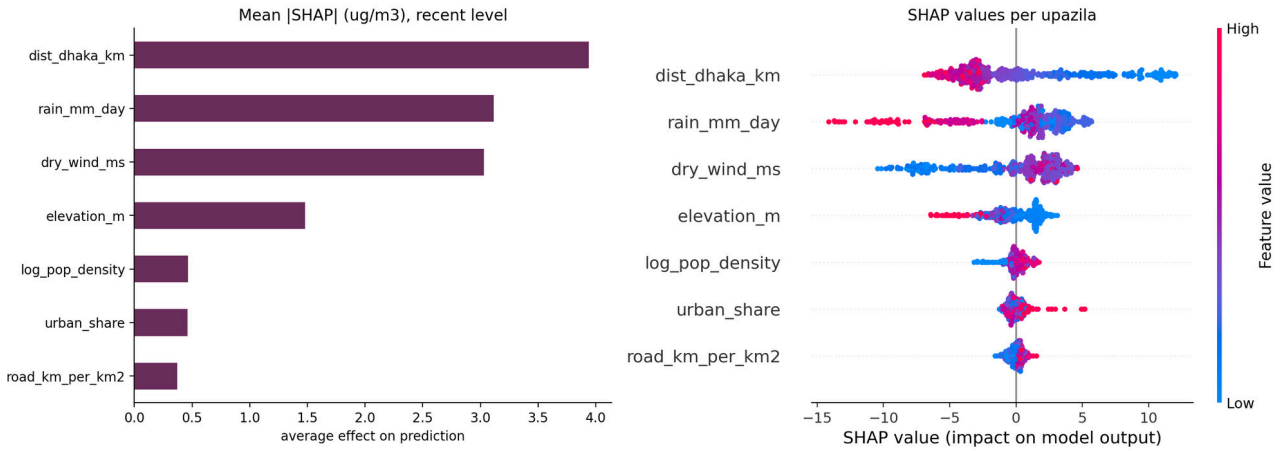


Figure 29. Mean absolute SHAP (left) and SHAP values per upazila (right) for the 2020-2024 level.

The SHAP dependence plot shows that distance to Dhaka adds up to 12 µg/m³ within about 50 km of Dhaka and takes about 5 µg/m³ off beyond 200 km. Higher dry-season wind goes with higher predicted PM2.5, the opposite of what dispersion alone would suggest. I read this as winter north-westerly transport from the Indo-Gangetic Plain, but it is an association. XGBoost and the spatial error model agree on distance to Dhaka. They disagree on rainfall and wind, which the SEM shows are mostly standing in for regional structure.

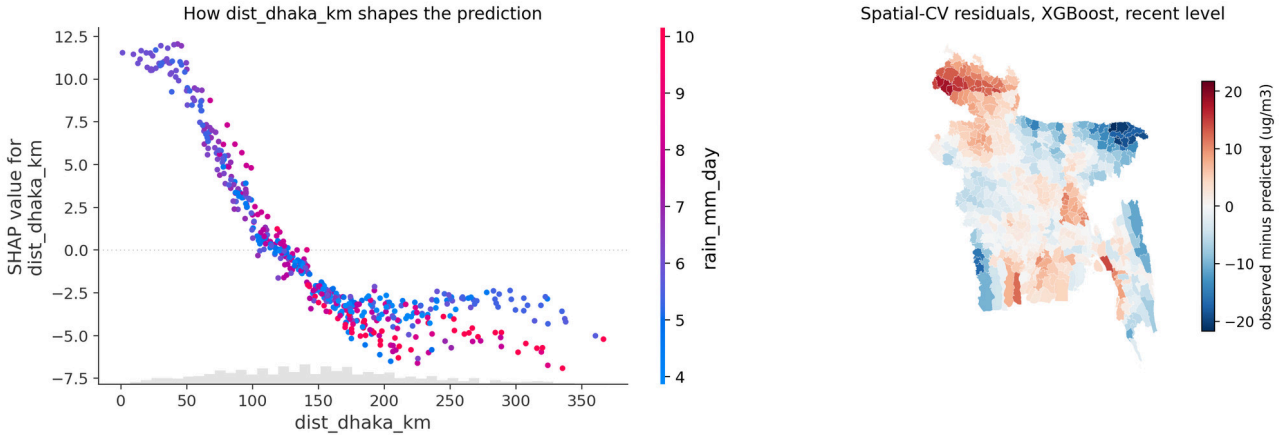


Figure 30. SHAP dependence for distance to Dhaka, coloured by rainfall (left), and spatial-CV residuals of XGBoost (right).

CHAPTER 11

What cleaner air would be worth

The AQLI relationship translates long-term PM2.5 above the WHO guideline into life expectancy: 0.98 years per 10 µg/m³. Applied to the 2020-2024 mean, the average person in Bangladesh loses an estimated 6.1 years of life expectancy, or about 1,033 million life-years across 168 million people. This is an estimate from a published relationship, not a count of deaths.

Table 16. Life expectancy lost and gained by division (AQLI method, 2020-2024).

Division	People (M)	PM2.5	Years lost vs WHO 5	Years gained at 35	Years gained at 25
Dhaka	45.1	79.7	7.3	4.4	5.4
Barishal	9.3	67.1	6.1	3.1	4.1
Chattogram	33.8	66.3	6.0	3.1	4.1
Rajshahi	20.7	65.9	6.0	3.0	4.0
Rangpur	17.9	64.8	5.9	2.9	3.9
Khulna	17.8	64.2	5.8	2.9	3.8
Mymensingh	12.5	60.2	5.4	2.5	3.5
Sylhet	11.2	44.8	3.9	1.0	1.9
Bangladesh	168.3	67.6	6.1	3.2	4.2

Meeting only the national standard of 35 µg/m³ would win back about 3.2 years per person on average, and 4.4 years in Dhaka division. That needs an average cut of 46% in PM2.5. Reaching the WHO guideline would need a 92% cut. The worst upazila, Savar, loses an estimated 7.9 years.

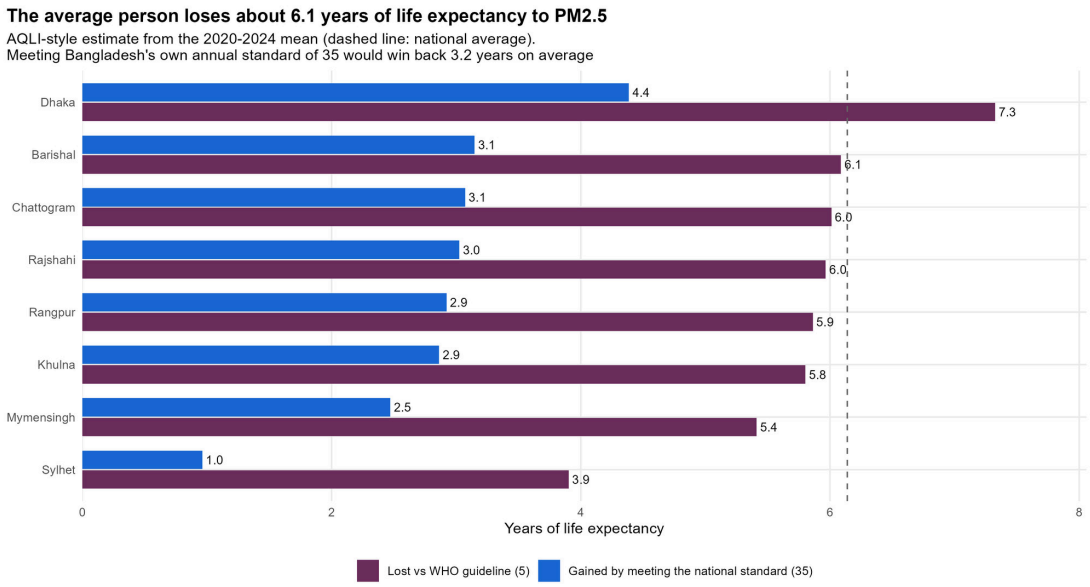


Figure 31. Years of life expectancy lost against the WHO guideline, and gained by meeting the national standard, by division.

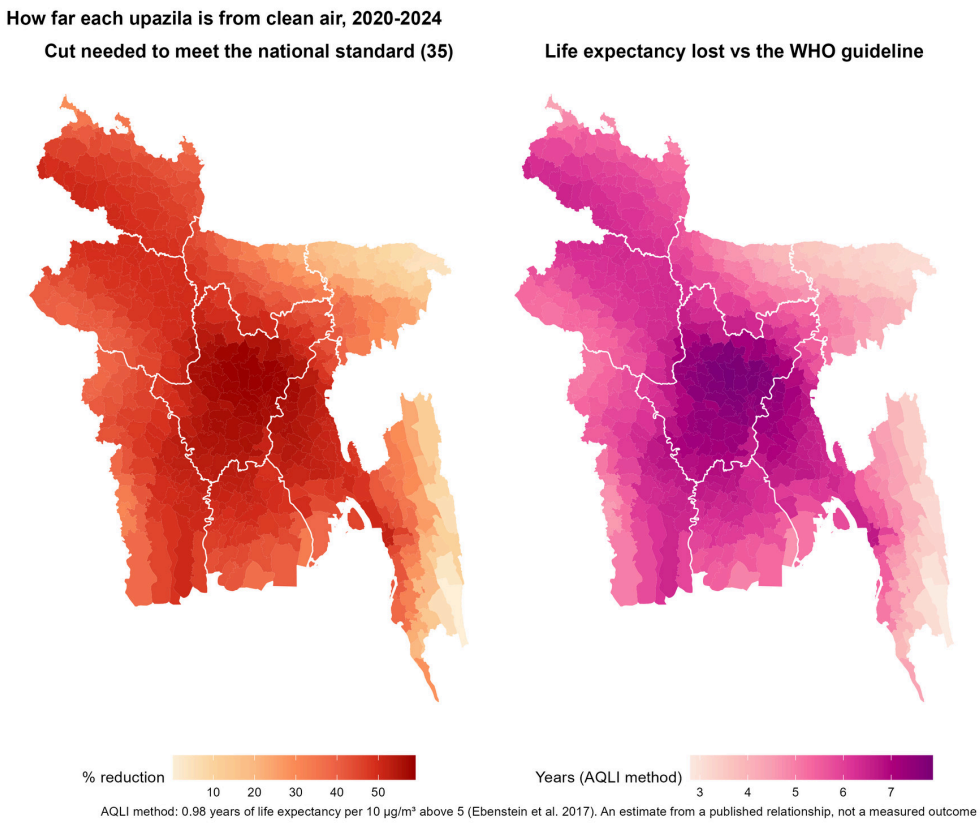


Figure 32. Cut needed to meet the national standard (left) and life expectancy lost against the WHO guideline (right), by upazila, 2020-2024.

CHAPTER 12

Power BI dashboard

The results are shown in a Power BI dashboard with seven report pages, a District profile drill-through and seven tooltip pages. The model has 25 tables from the R export, 21 relationships and about 120 measures. The measures only look up and format results computed in Python and R, so every number on the dashboard matches the notebooks and scripts. The maps use TopoJSON files for the 64 districts and 507 upazilas built from the HDX boundaries, with fixed colour scales so years can be compared.

Table 17. Dashboard pages.

Page	Question it answers
Executive summary	How bad is the air most people breathe, and is it getting worse?
Exposure map	Where is the air worst, and how has that changed?
People at risk	How many people live above each WHO target, and what would cleaner air be worth?
Trends	How fast did PM2.5 rise, and has it levelled off?
Seasonality	Which months are worst, and is the winter peak changing?
Hotspots	Where does pollution cluster, and how far does it spread?
Drivers	What goes with higher PM2.5, and does machine learning agree when tested honestly?
District profile	Drill-through for one district: trend, months, bands and upazilas

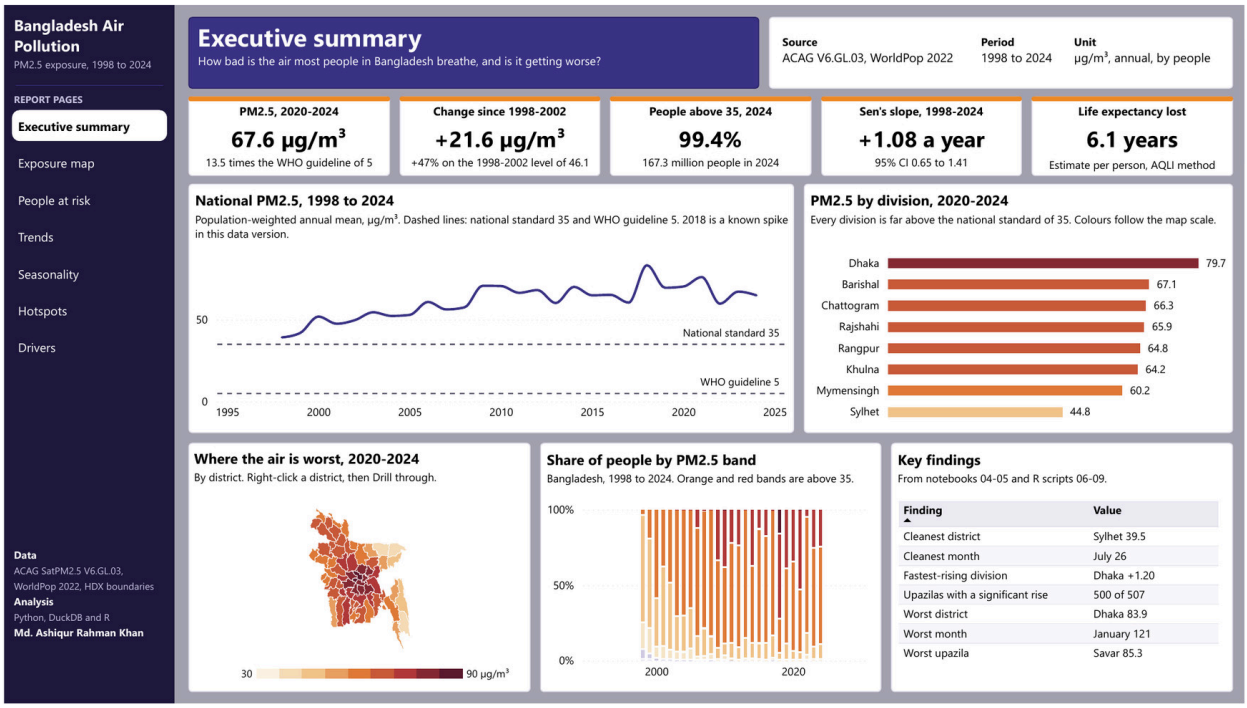


Figure 33. Dashboard: Executive summary.

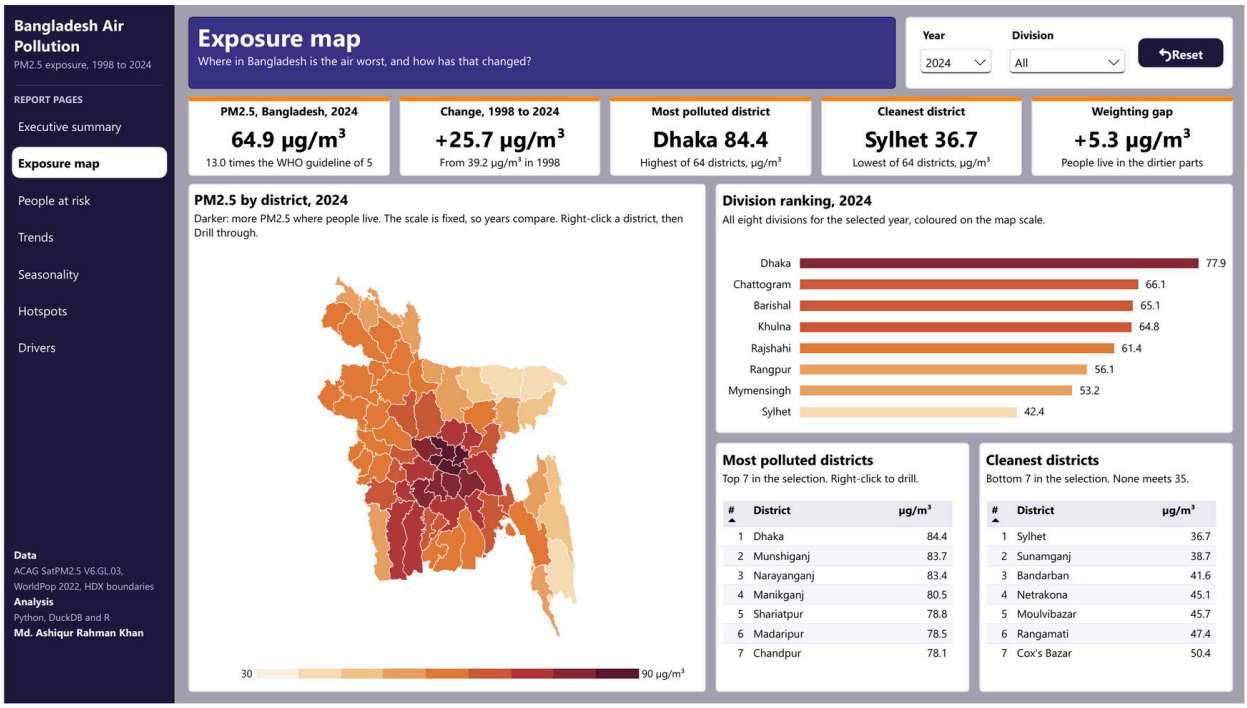


Figure 34. Dashboard: Exposure map.

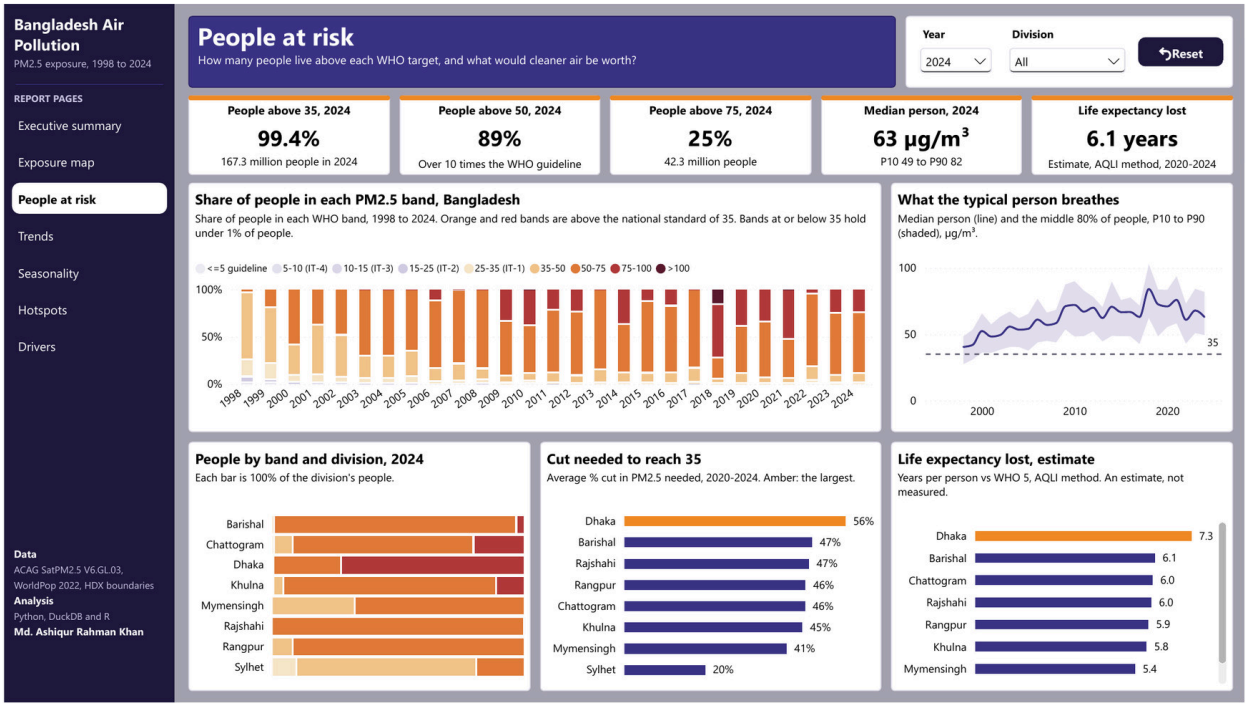


Figure 35. Dashboard: People at risk.

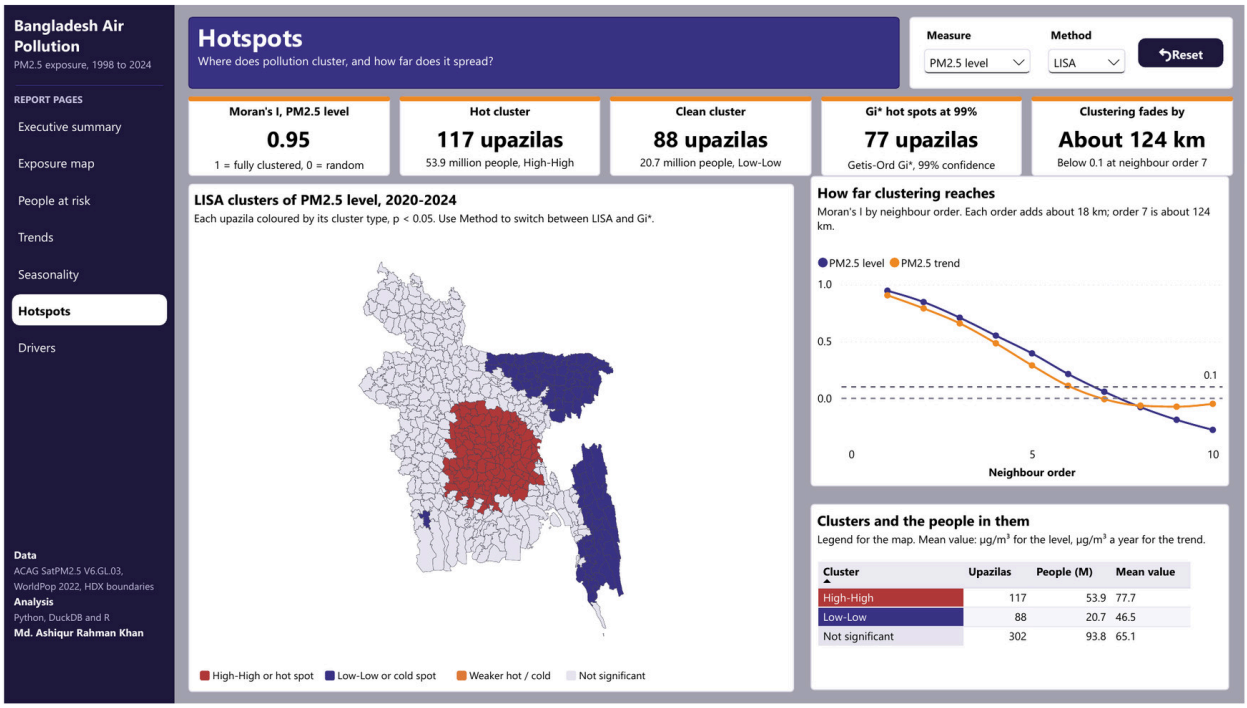


Figure 36. Dashboard: Hotspots.

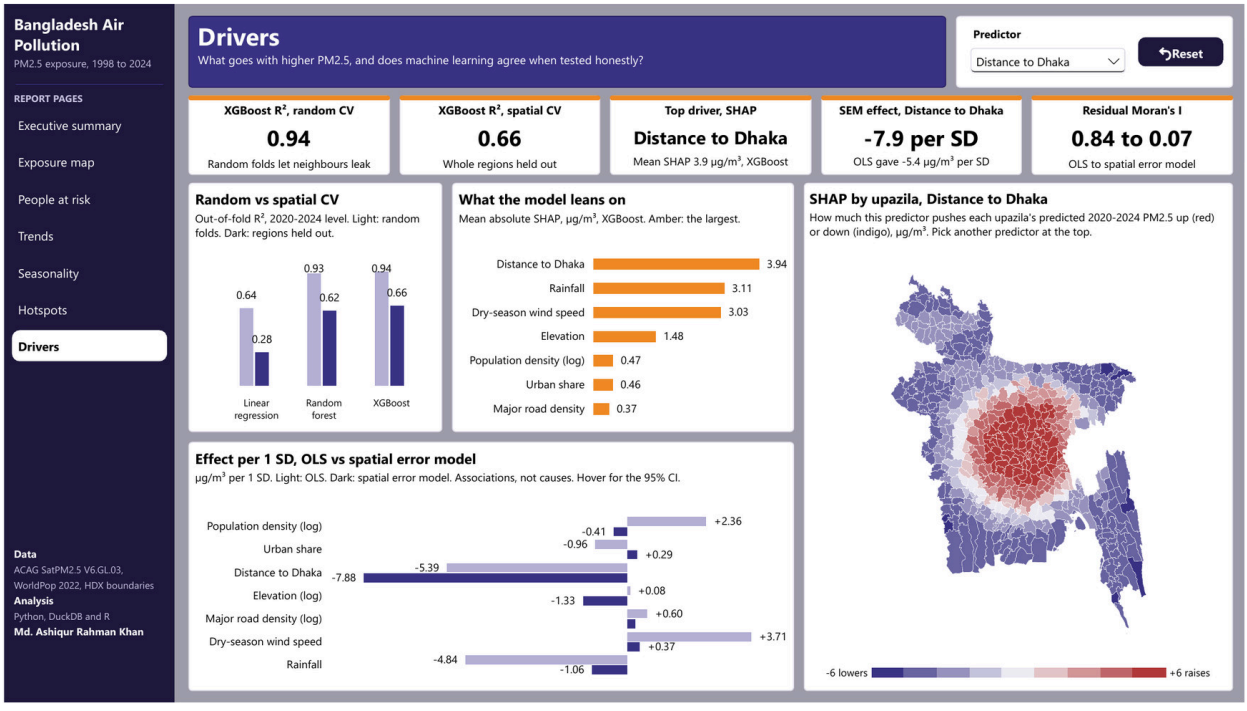


Figure 37. Dashboard: Drivers.

CHAPTER 13

Limitations

- **Modelled, not measured.** ACAG combines satellite aerosol data, a chemical transport model and ground monitors. At about 1 km it is smoother than reality and misses street-level peaks near roads and kilns. All values are estimated outdoor exposure.
- **Fixed population.** WorldPop 2022 weights every year. The trend shows how pollution changed where people live now, not exactly what people breathed in 1998. Migration into the Dhaka area means the true historical exposure was probably a little lower.
- **The 2018 release spike.** The 2018 value in V6.GL.03 is about 28% above its neighbours. It was kept and flagged. Sen's slope barely changes without it, but single-year values for 2018 should not be read on their own.
- **Two grids.** Monthly data is on a 0.1° grid. For small coastal upazilas the monthly average can differ from the annual value by up to 11 $\mu g/m^3$, so seasonality is reported only for divisions and districts.
- **Areal associations.** The regression and SHAP results describe upazilas, not people, and they are not causal. Brick kilns, cross-border transport and industry are not measured directly, so distance to Dhaka is partly standing in for them.
- **Smooth surface.** The very high Moran's I partly reflects the smoothness of the satellite surface. The LISA and Gi* maps are reliable at the regional scale, not for a single upazila boundary.
- **Life expectancy is an estimate.** The AQLI figures come from a published exposure-response relationship from China. They show the size of the problem, not measured deaths, and they assume the relationship holds in Bangladesh.

CHAPTER 14

Conclusion

Almost everyone in Bangladesh breathes air far above every WHO target. The average person was exposed to 67.6 μg/m³ in 2020-2024, 47% more than in 1998-2002, and 99.4% of people were above the national standard in 2024. Exposure rose quickly until about 2010 and has stayed at that high level since. Dhaka and the districts around it are the worst, but the fastest rises in the last 27 years were in the north-west and on the Khulna coast, and the monsoon months got worse faster than winter.

The strongest pattern in the data is regional. Distance to Dhaka holds up in both the spatial error model and XGBoost, while rainfall and wind mostly describe where a place is. The model checks mattered: random cross-validation and plain OLS both made the covariates look more powerful than they are. Meeting the national standard alone would add an estimated 3.2 years of life expectancy for the average person.

The next steps would be to add brick kiln locations and cross-border transport as covariates, to test the trend with a historical population series, and to compare the satellite values with the government's monitoring stations where they exist.

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Appendix A. National yearly series

Table 18. National PM2.5 by year (µg/m³).

Year	Population-weighted	Area-weighted	Times WHO	Year	Population-weighted	Area-weighted	Times WHO
1998	39.2	36.2	7.8	2012	68.1	64.3	13.6
1999	41.9	38.4	8.4	2013	60.2	55.8	12.0
2000	51.9	47.3	10.4	2014	70.1	64.2	14.0
2001	47.6	43.9	9.5	2015	64.9	60.6	13.0
2002	49.8	45.6	10.0	2016	65.2	61.0	13.0
2003	54.6	50.2	10.9	2017	60.5	55.9	12.1
2004	52.3	48.5	10.5	2018	83.3	77.0	16.7
2005	52.9	48.1	10.6	2019	69.5	64.2	13.9
2006	60.9	55.7	12.2	2020	70.3	64.9	14.1
2007	56.3	52.4	11.3	2021	76.1	69.6	15.2
2008	57.5	53.3	11.5	2022	59.8	55.0	12.0
2009	70.7	64.8	14.1	2023	67.2	61.3	13.4
2010	70.6	64.4	14.1	2024	64.9	59.6	13.0
2011	66.4	60.1	13.3				

Appendix B. Files and reproducibility

All code, processed data, outputs and the dashboard are in the GitHub repository Ashiq-r-khan/Bangladesh-Air-Pollution-Exposure-Analysis. Notebooks 01 to 05 run in Google Colab with the project folder in Google Drive, in order. R scripts 06 to 10 run in RStudio from the project folder, in order. The raw downloads are not stored in the repository; notebook 01 recreates them from the sources listed in *data/raw/download_manifest.csv*. The 27 tables behind the dashboard are in *outputs/powerbi*, with a table dictionary and the relationships file.